

Large Language Models

Understanding In-Context Learning

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In-Context Learning



In-Context Learning (ICL)

- Given a **prompt** including:
 - An optional description of the task
 - Few-shot training examples in a prompt format demonstrating the task
 - The test input

“Apple → Red, Lime → Green, Corn →”

“Albert Einstein was German \n Mahatma Gandhi was Indian \n Marie Curie was”

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 cheese => ..... ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← example
3 cheese => ..... ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

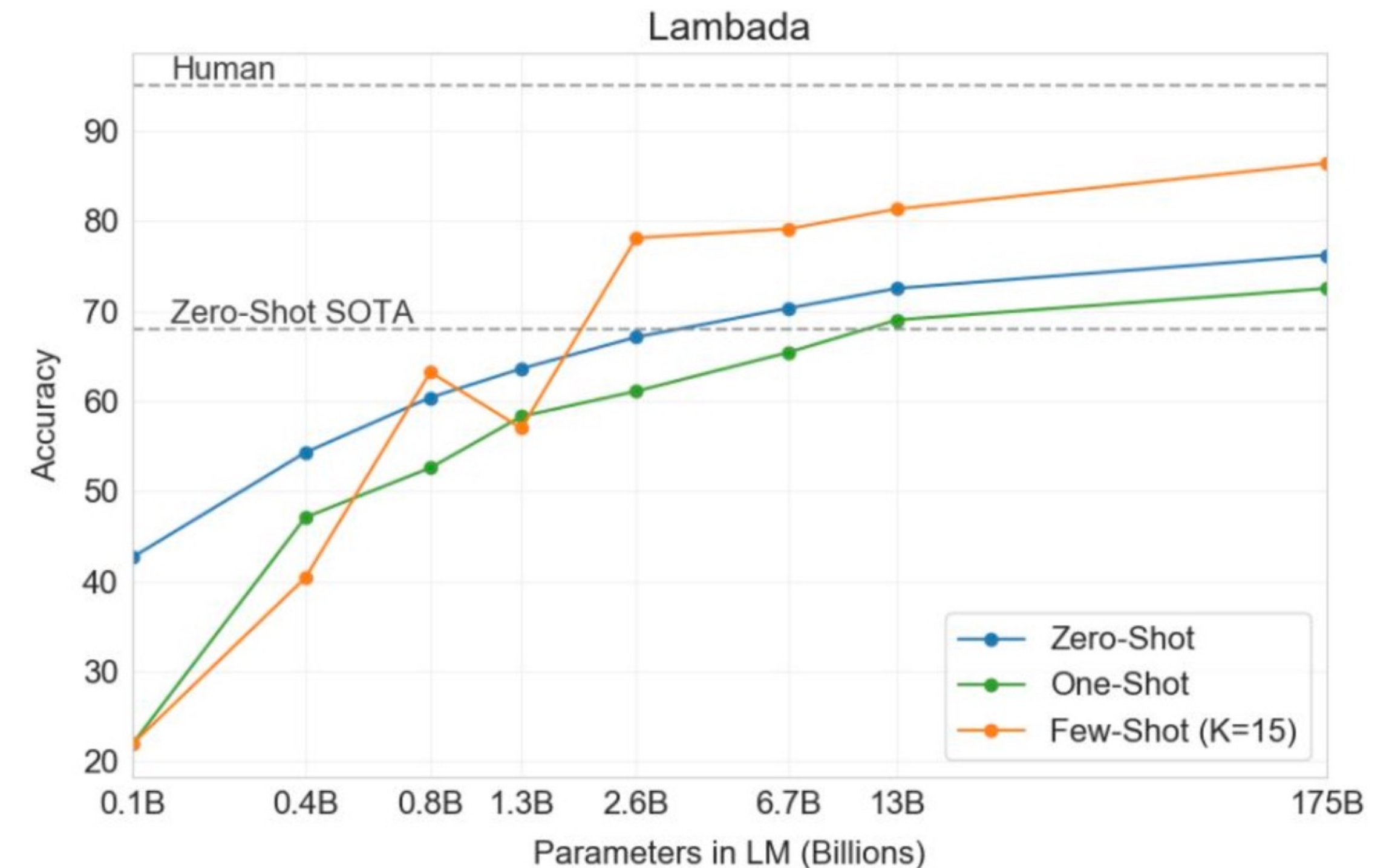
```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivrée ←
4 plush girafe => girafe peluche ←
5 cheese => ..... ← 3/71 prompt
```

What can ICL do?

- No parameter tuning is required
- Only need few examples for downstream tasks
- GPT-3 improved SOTA on LAMBADA by 18%!



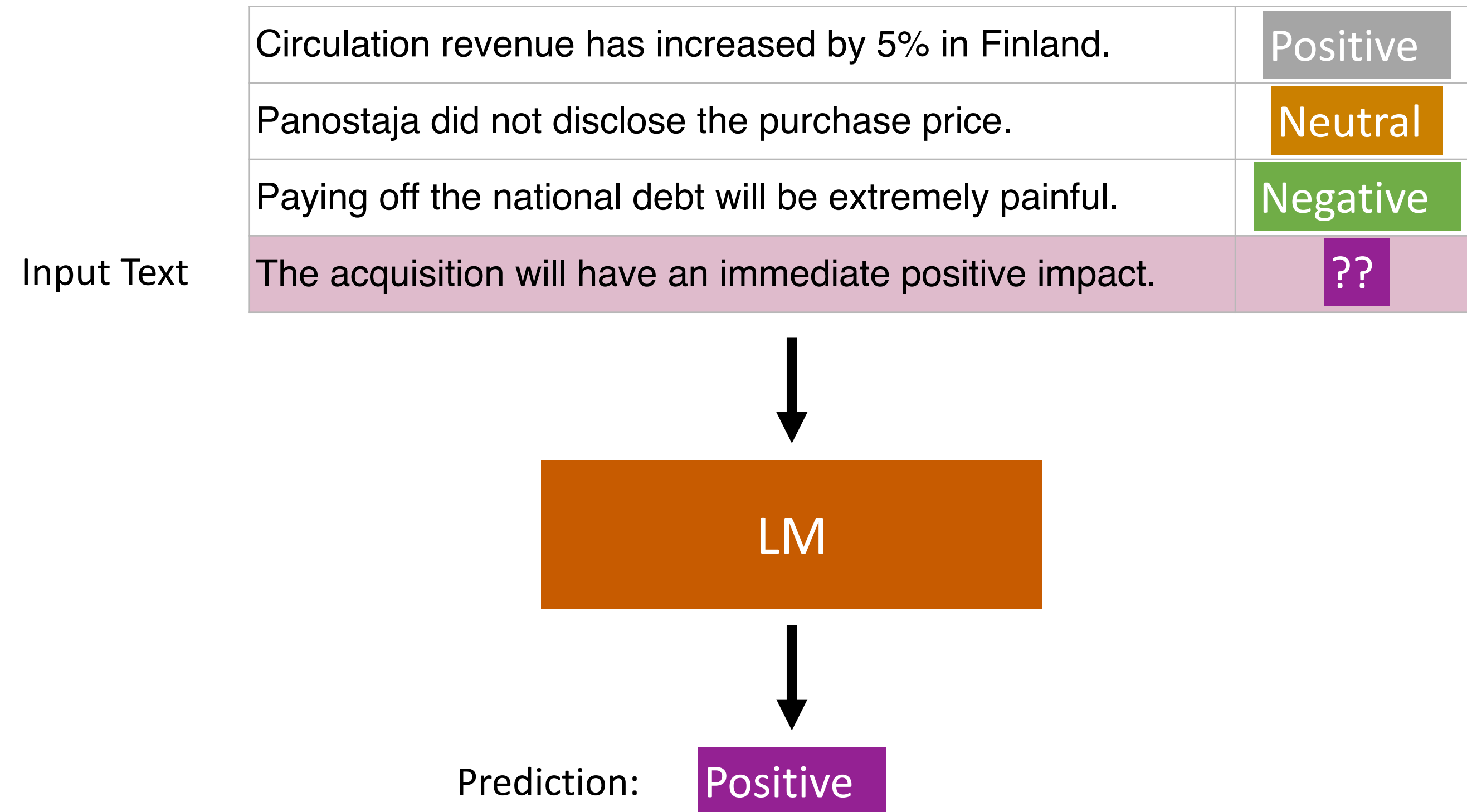
Works like magic!



ICL: Advantages

- enables rapid prototyping
- provides a fully natural language interface
- reuses the same model for each task
 - reduces memory requirements and system complexity when serving many different tasks.
- Finetuning can be unstable in the few-shot setting (Schick & Schutze, 2021)

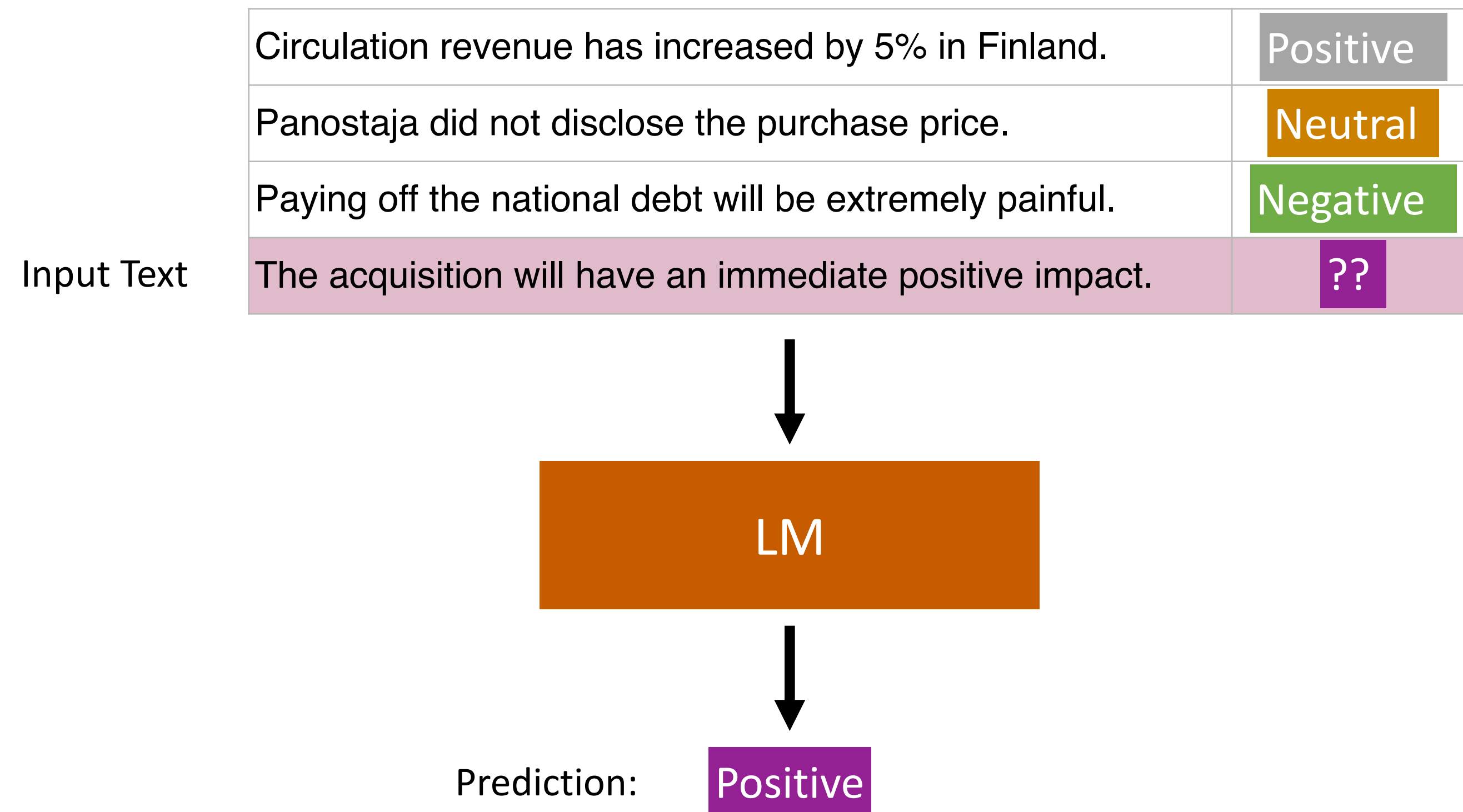
We don't know how models in-context learn



Note

Learns to do a downstream task by conditioning on input-output **examples**

We don't know how models in-context learn



Note

No weight update and model is not explicitly pre-trained to learn from examples

Question

How does it know what to do then?



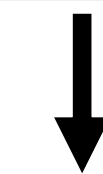
- Circulation revenue has increased by 5% in Finland. Positive
- Panostaja did not disclose the purchase price. Neutral
- Paying off the national debt will be extremely painful. Negative
- The company anticipated its operating profit to improve. ??

- Circulation revenue has increased by 5% in Finland. Finance
- Panostaja did not disclose the purchase price. Sports
- Paying off the national debt will be extremely painful. Tech
- The company anticipated its operating profit to improve. ??

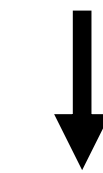
Understanding in-context learning

- A mathematical framework (Xie et al., 2022)
 - **Bayesian inference** view: understand how in-context learning emerges
- Empirical evidence (Hendel et al., 2023)
 - ICL creates task vectors
- Empirical evidence (Min et al., 2022)
 - Which aspects of the prompt affect downstream task performance?

- Circulation revenue has increased by 5% in Finland. Positive
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LM



Prediction: Positive

- Circulation revenue has increased by 5% in Finland. Finance
- Panostaja did not disclose the purchase price. Sports
- Paying off the national debt will be extremely painful. Tech
- The company anticipated its operating profit to improve. ??

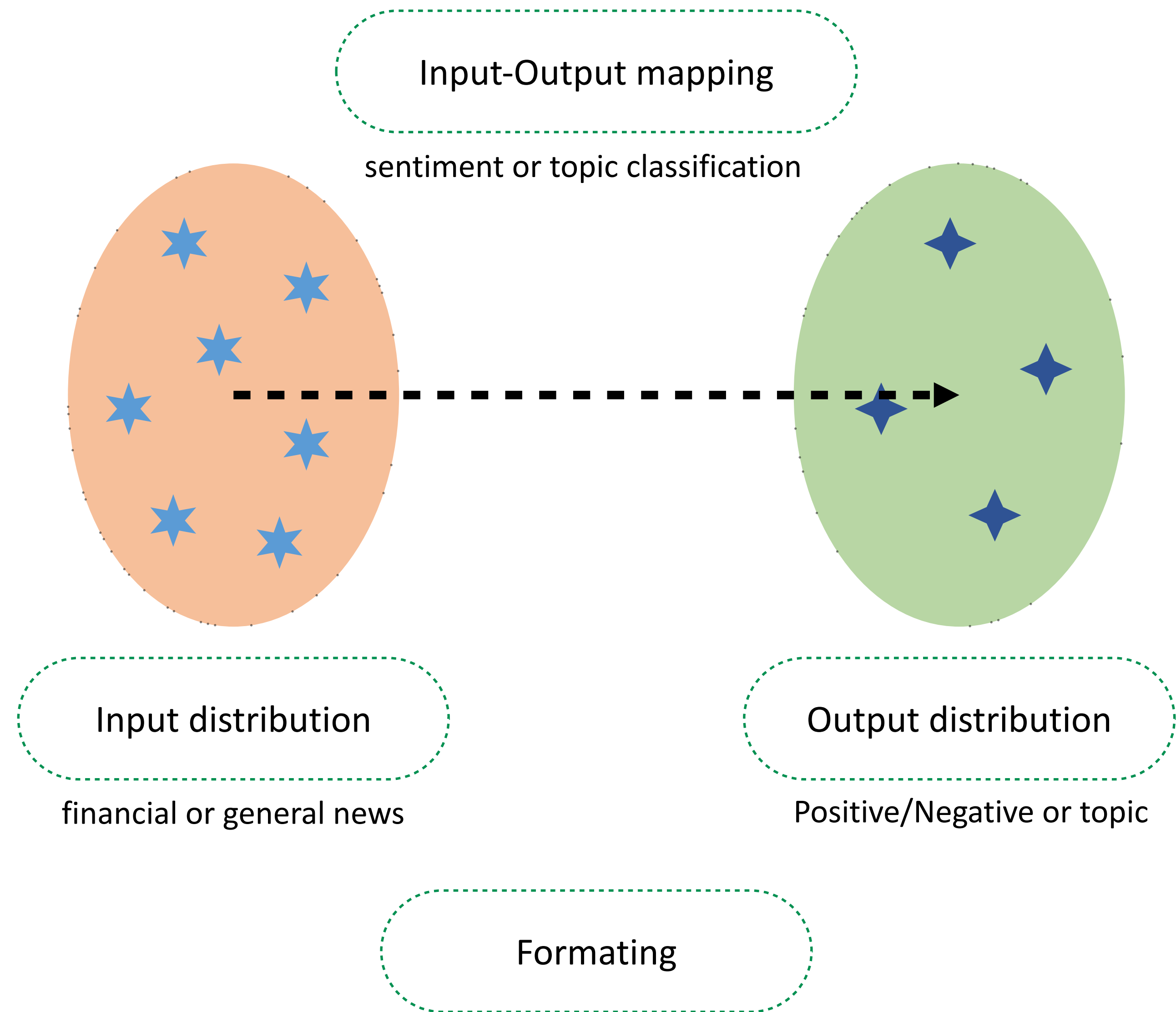


LM



Prediction: Finance

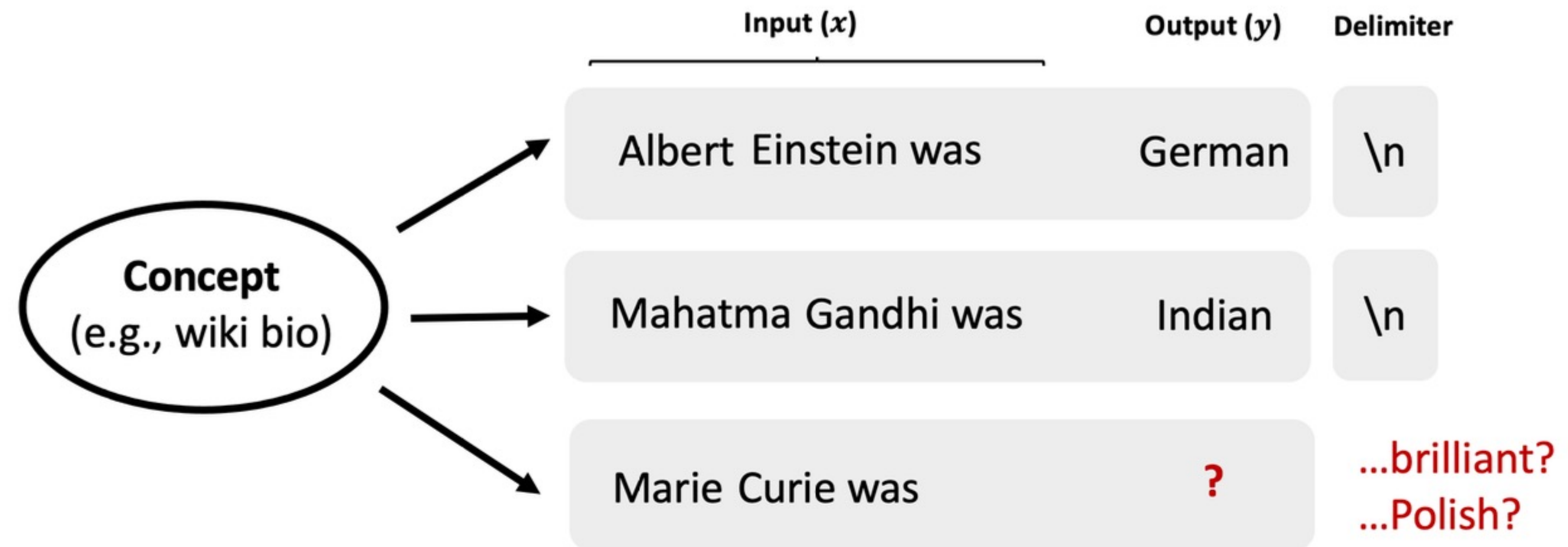
Model Requirements



1. **Pretraining documents** are conditioned on a **latent concept** (e.g., biographical text)



2. Create **independent examples** from a **shared concept**. If we focus on full names, wiki bios tend to relate them to nationalities.

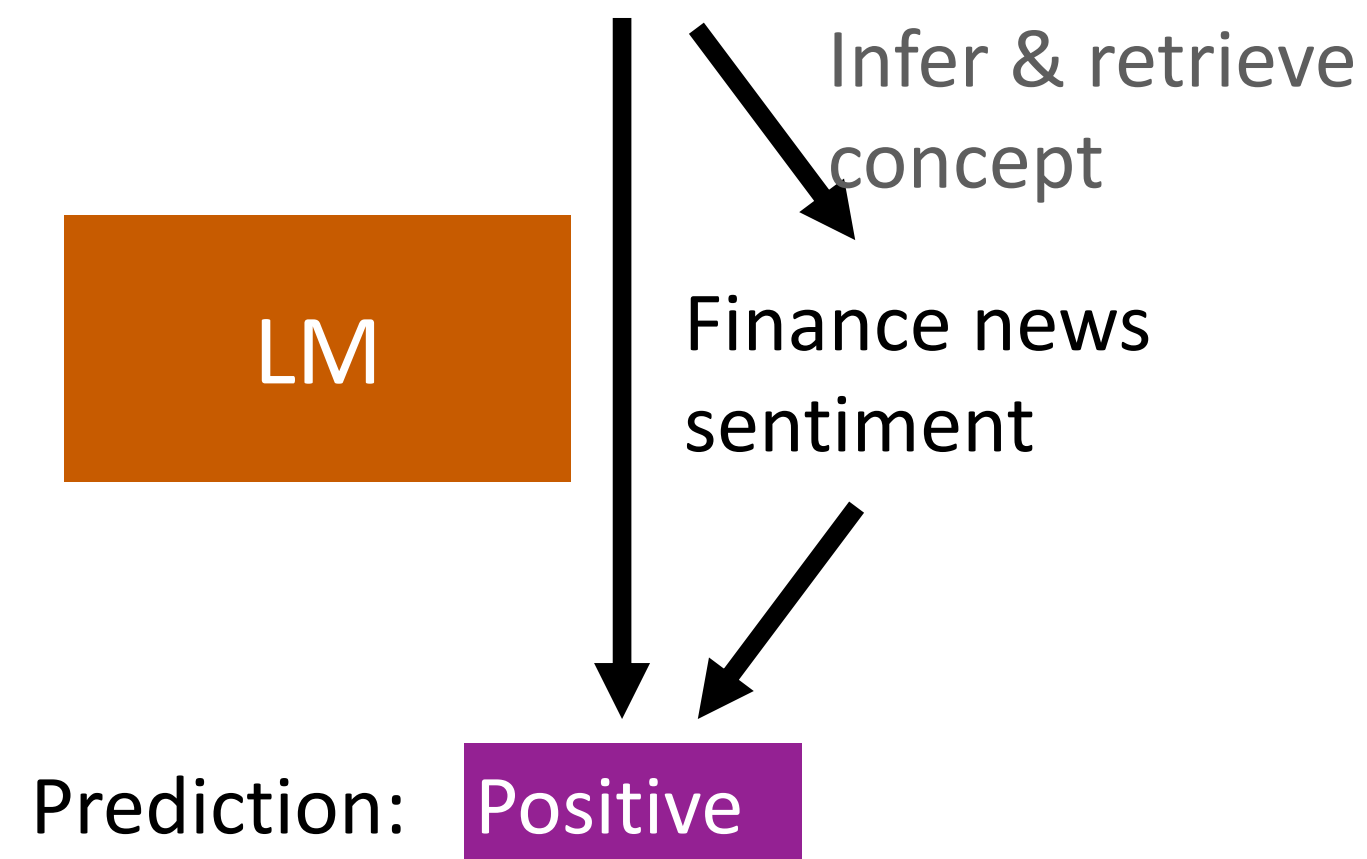


3. **Concatenate examples into a prompt** and predict next word(s). **Language model (LM)** implicitly **infers the shared concept** across examples despite the unnatural concatenation

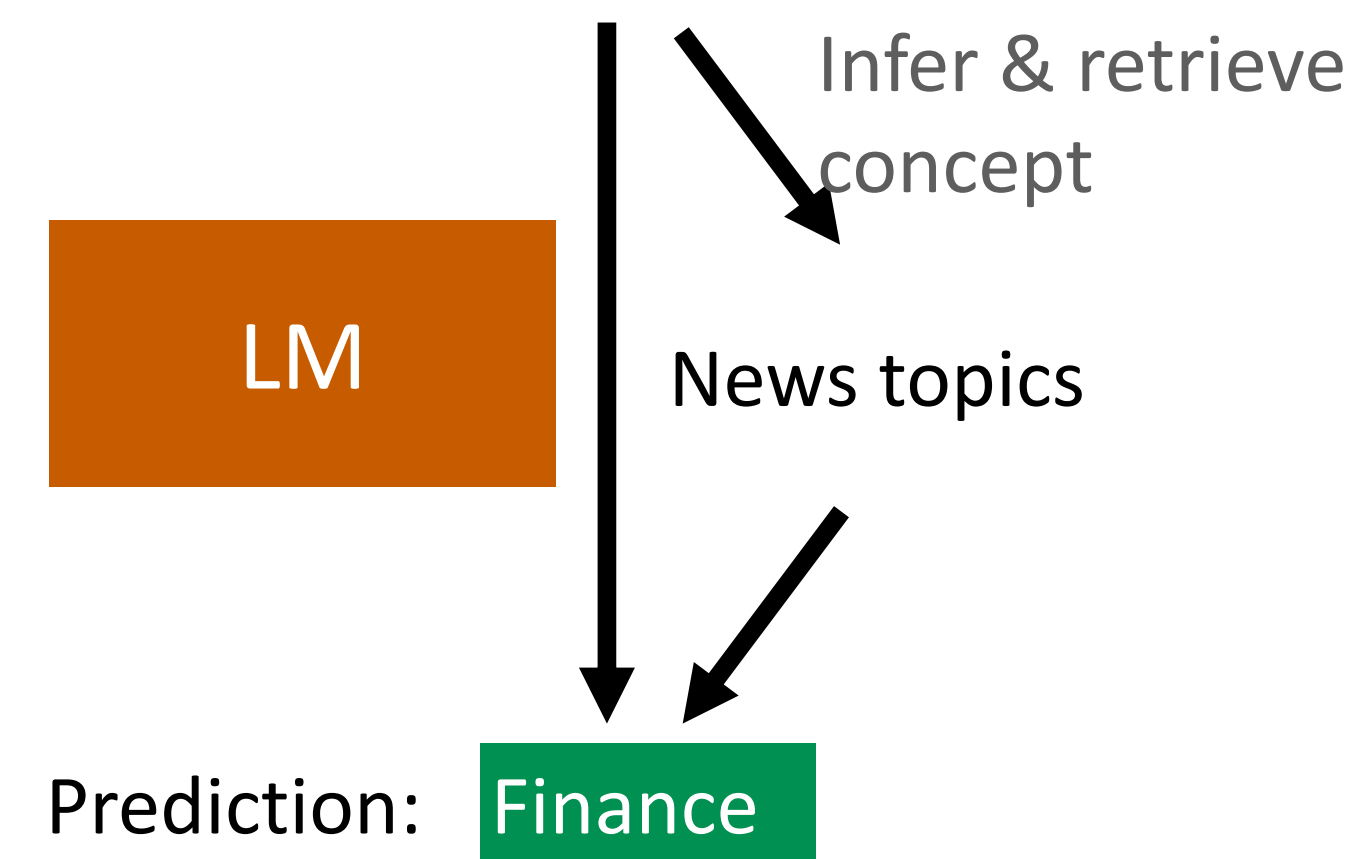


Concepts (long-term coherence)

- Circulation revenue has increased by 5% in Finland. **Positive**
- Panostaja did not disclose the purchase price. **Neutral**
- Paying off the national debt will be extremely painful. **Negative**
- The company anticipated its operating profit to improve. **??**



- Circulation revenue has increased by 5% in Finland. **Finance**
- Panostaja did not disclose the purchase price. **Sports**
- Paying off the national debt will be extremely painful. **Tech**
- The company anticipated its operating profit to improve. **??**



Note

A latent variable that contains various **document-level statistics**: a distribution of words, a format, a relation between sentences, and other semantic and syntactic relations in general.

Hypothesis

Maximum likelihood $\sim$! ; ! Bayesian inference ; $\Leftarrow$ قسری



Language model (LM) uses the in-context learning prompt to “locate” a previously learned concept to do the in-context learning task

Bayesian inference!

$$p(\text{output}|\text{prompt}) = \int_{\text{concept}} p(\text{output}|\text{concept}, \text{prompt})p(\text{concept}|\text{prompt})d(\text{concept})$$

How does the LM learn to do Bayesian inference?

- **Pre-train:** To predict the next token during pre-training, the LM must infer the latent concept for the document using evidence from the previous sentences.
- **In-context learning:** If the LM also infers the prompt concept using demonstrations in the prompt, then in-context learning succeeds!



Mixture of Hidden Markov Models (HMM)

Pre-training distribution

- Each pre-training document is a length T sequence sampled by

$$p(o_1, \dots, o_T) = \int_{\theta \in \Theta} p(o_1, \dots, o_T | \theta) p(\theta) d\theta$$

where θ is a family of concepts that defines a distribution over observed tokens o

- Assumption: $p(o_1, \dots, o_T | \theta)$ is defined by a Hidden Markov Model (HMM). The concept θ determines the transition probability matrix of the HMM hidden states $h_1, \dots, h_T$ from a hidden state set H .

Pre-training distribution

- If the pre-training data is a mixed of finance news **sentiment** task and news **topics** task, intuitively, we could say there are two concepts θ_1 and θ_2 .
- $p(\text{Paying off the national debt will be extremely painful}) =$
 $\frac{1}{2} p(\text{Paying off the national debt will be extremely painful} \mid \theta_1)$
+
 $\frac{1}{2} p(\text{Paying off the national debt will be extremely painful} \mid \theta_2)$

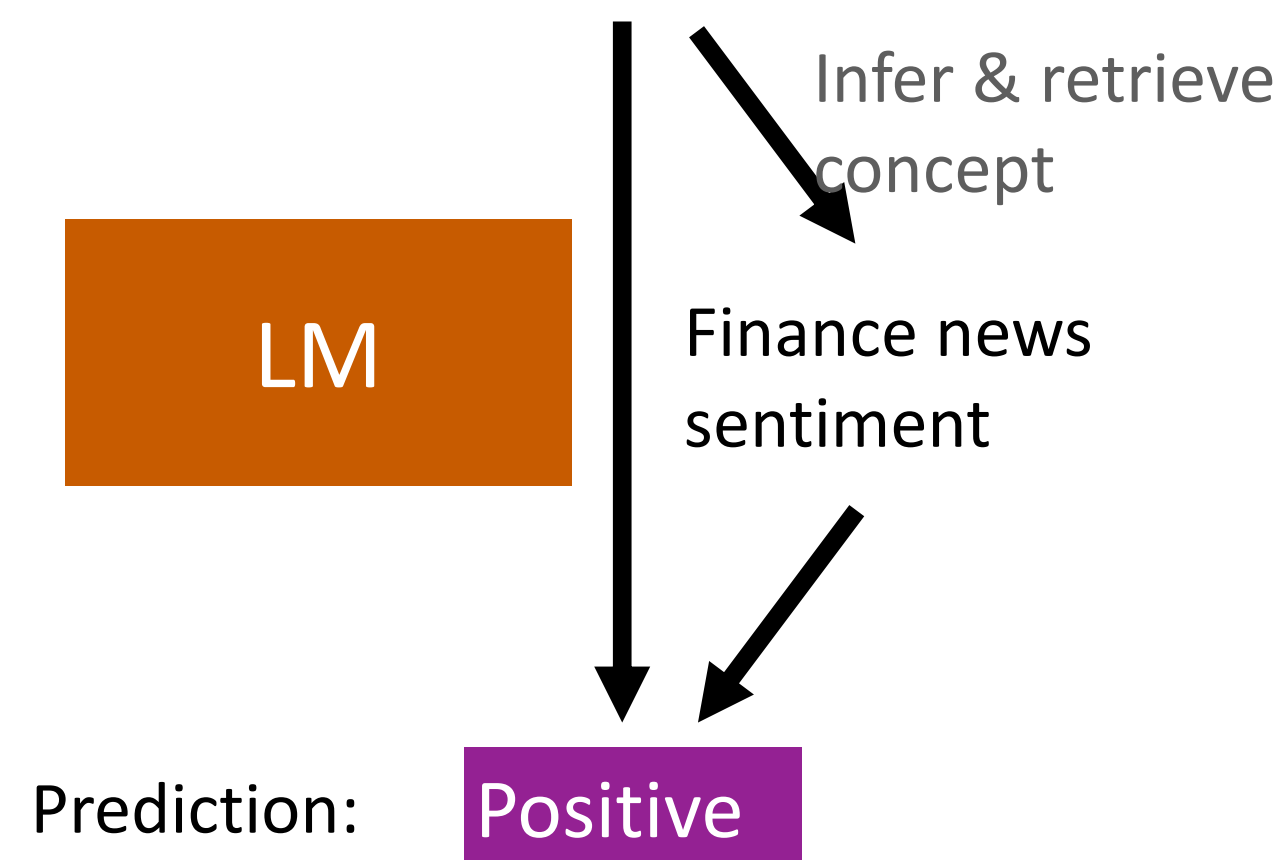
Prompt distribution

- The prompt is a sequence of demonstrations S_n followed by the test example x_{test} :

$$[S_n, x_{test}] = [x_1, y_1, o^{delim}, x_2, y_2, o^{delim}, \dots, x_n, y_n, o^{delim}, x_{test}]$$

- Given $\theta^* \in \Theta$, the prompt is a concatenation of n independent demonstrations and **1** test input x_{test} that are all conditioned on θ^* .

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[Circulation revenue has increased by 5% in Finland., Positive,
 #,
 Panostaja did not disclose the purchase price., Neutral,
 #,
 Paying off the national debt will be extremely painful., Negative,
 #,
 The company anticipated its operating profit to improve.]

Under some assumptions, as $n \rightarrow \infty$,

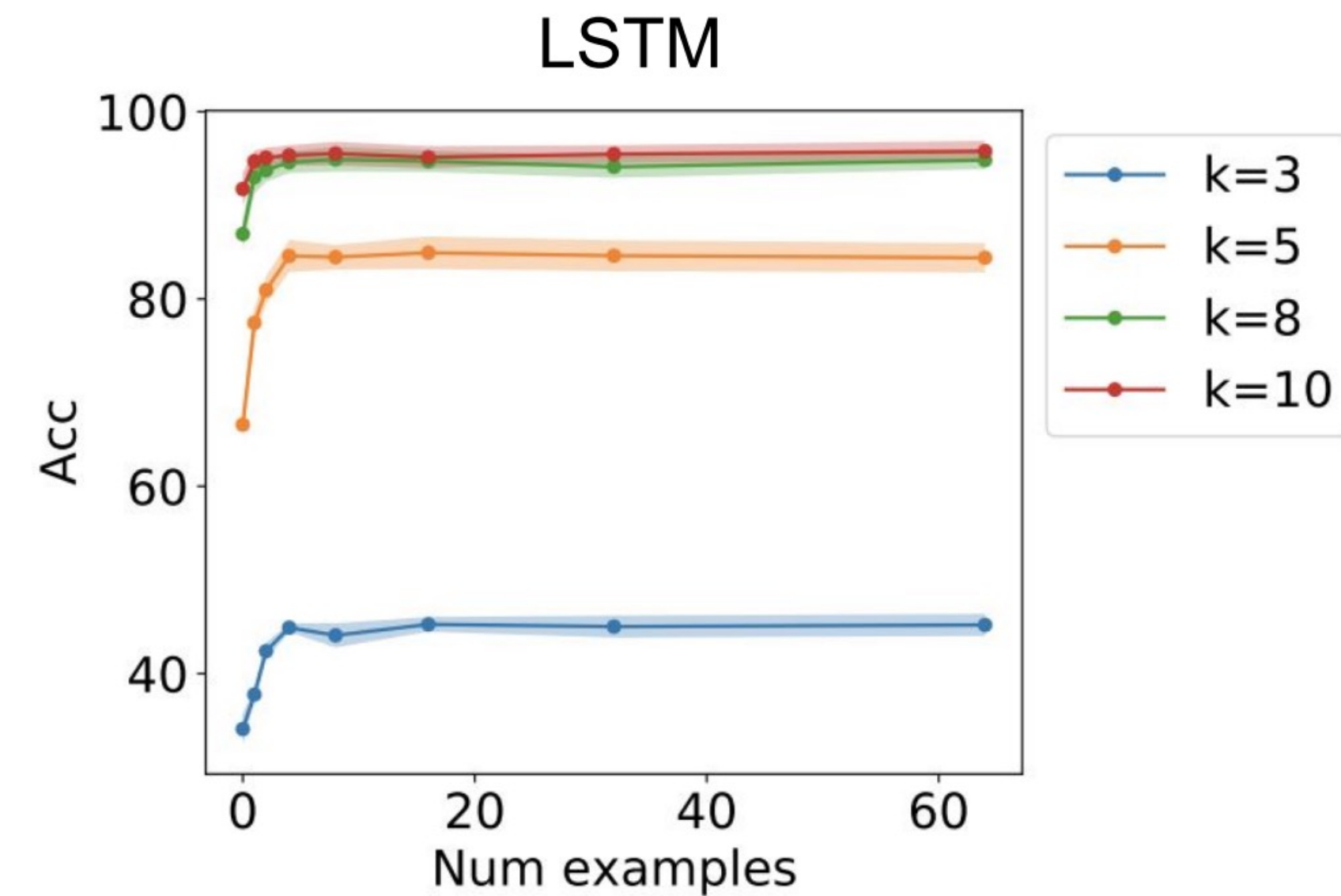
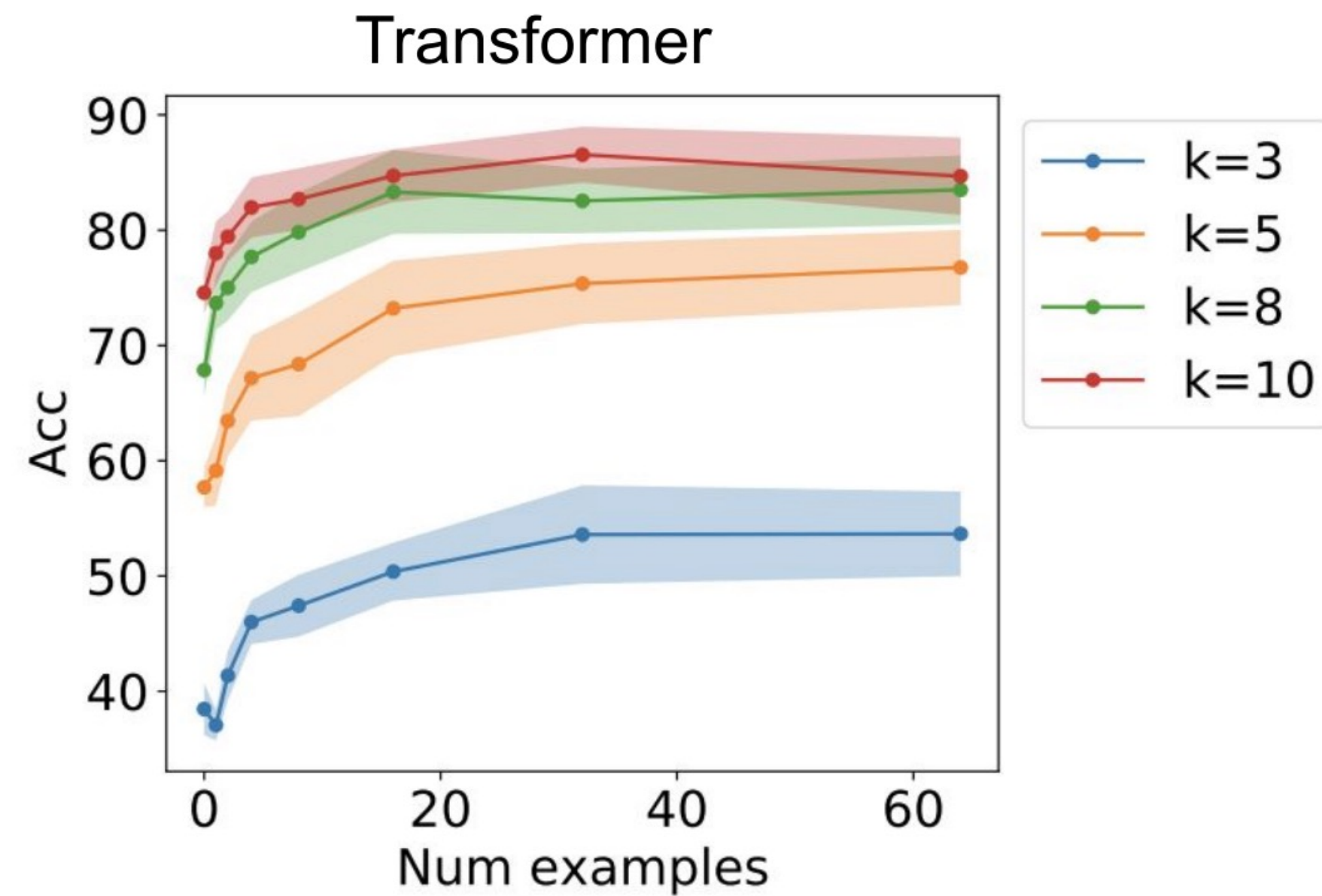
$$\operatorname{argmax}_y p(y|S_n, X_{test}) \rightarrow \operatorname{argmax}_y p_{prompt}(y|X_{test})$$

- $p_{prompt} \sim p(\cdot | \theta^*)$
- The in-context predictor asymptotically achieves the optimal expected error
- More examples $\rightarrow$ More signals for Bayesian inference $\rightarrow$ Smaller error

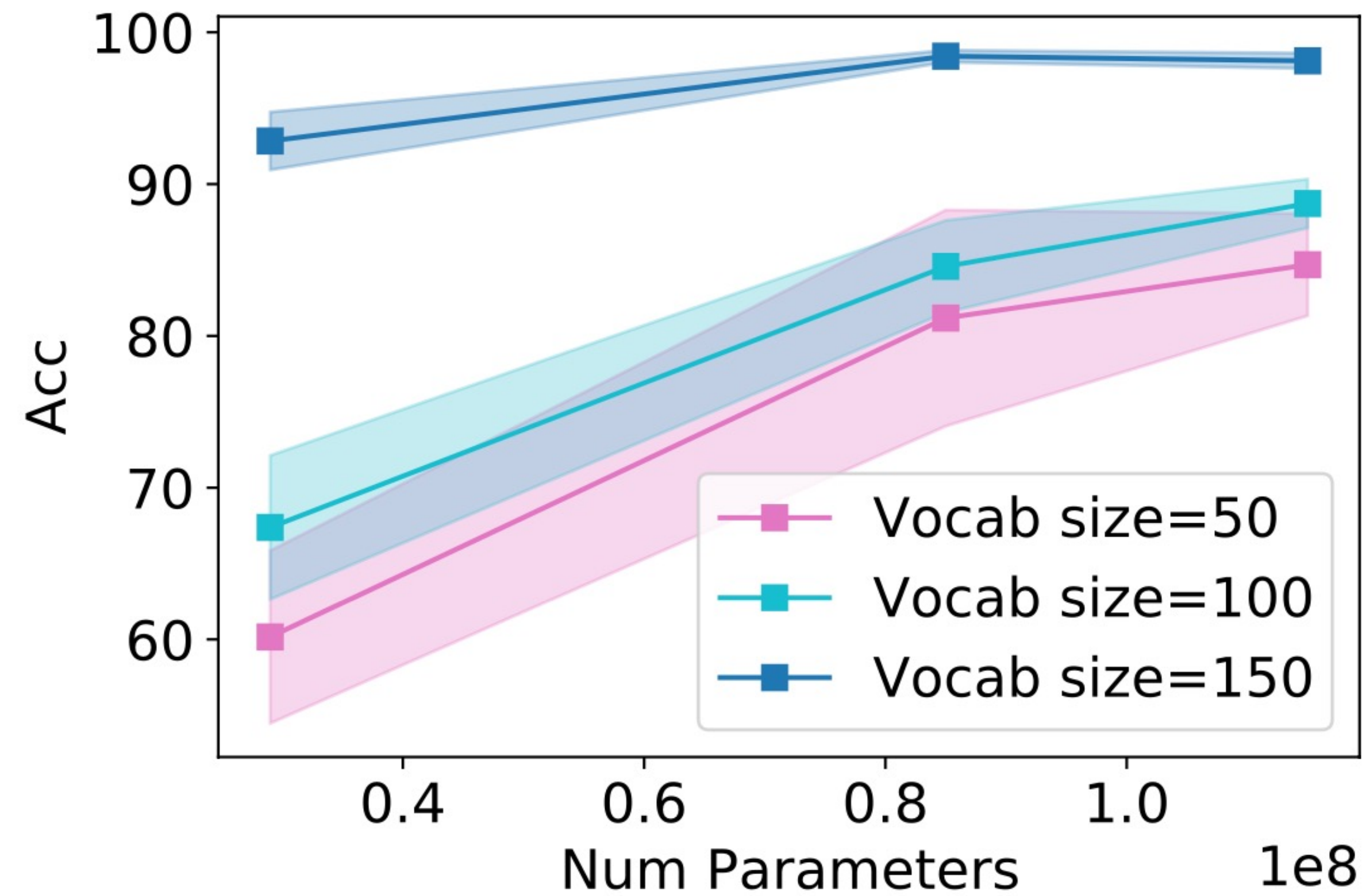
GINC: Generative In-Context learning Dataset

- A synthetic pretraining dataset and in-context learning testbed with the latent concept structure.
- **Pre-training:** a uniform mixture of HMMs over a family of 5 concepts, 1000 pre-training documents, ~10 million tokens in total
- **Prompts:** 0~64 training examples, example length $k=3, 5, 8, 10$
- GPT-2-based Transformers and LSTMs
- Vocabulary size: 50, 100, 150

اثرایشی نمونه از به جای به به $\sim$ long in-context learning برهان
fine-tuning

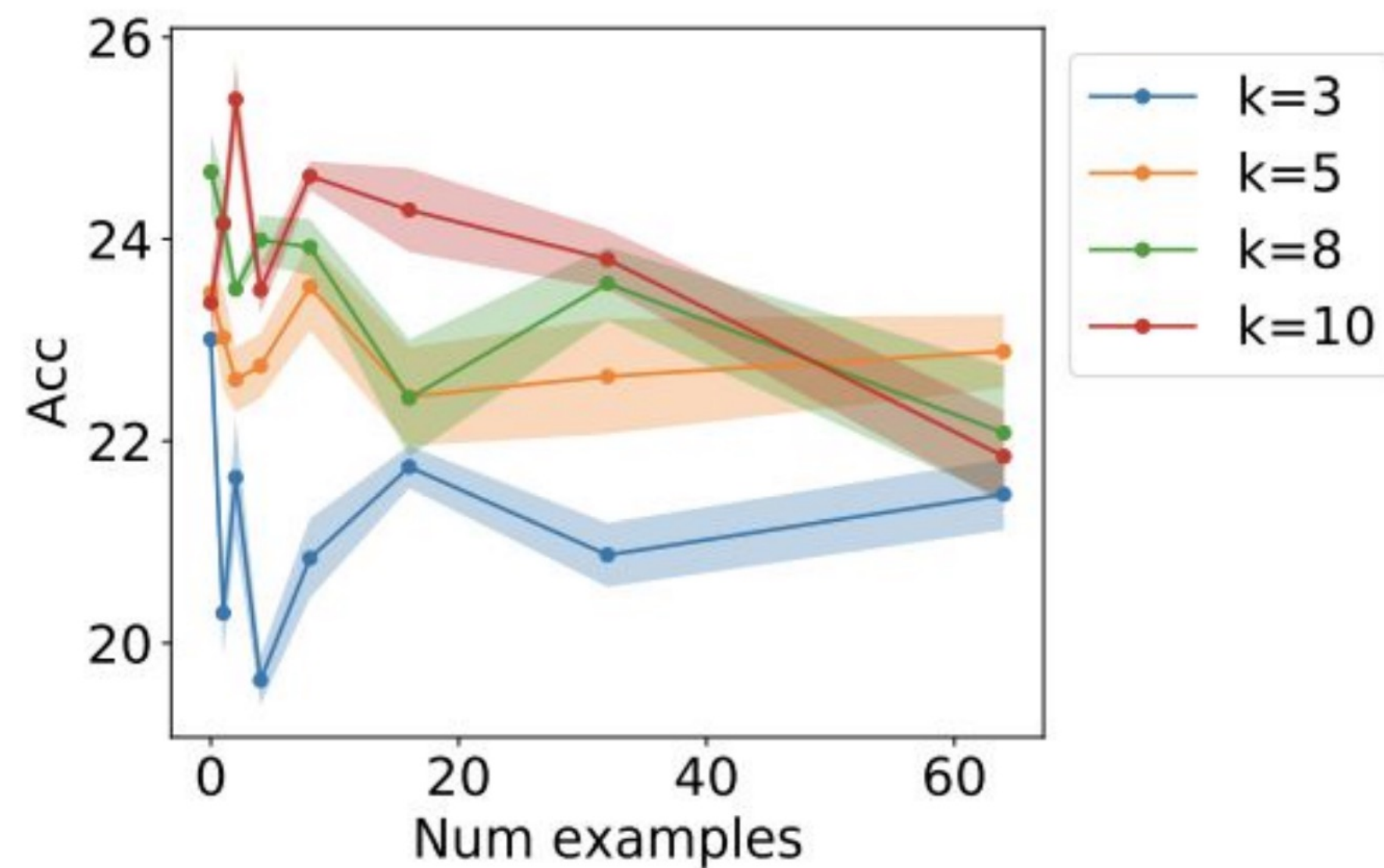


Accuracy increases with number of examples n and length of each example k , which is consistent with the theoretical results.

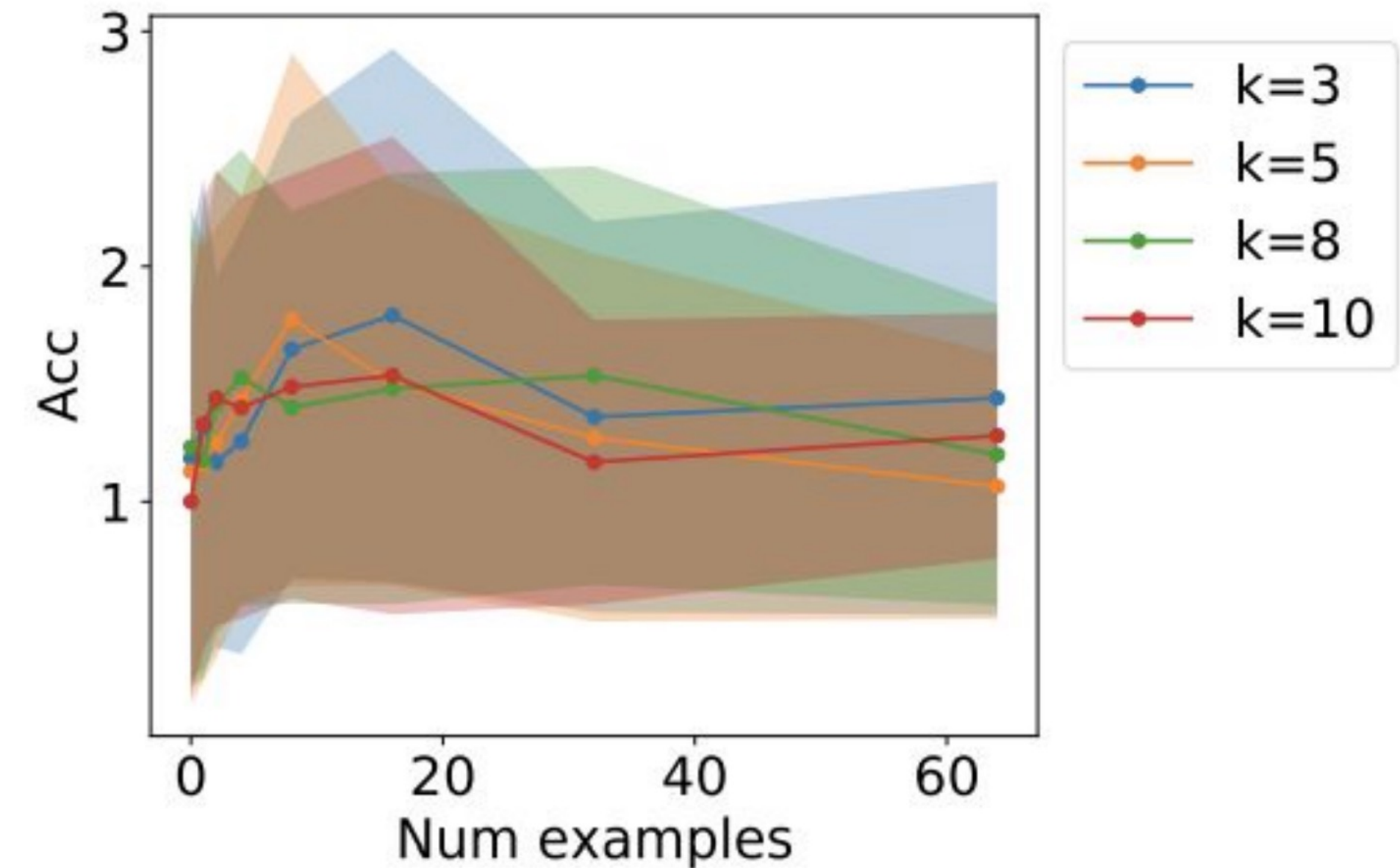


In-context accuracy (95% intervals) of Transformers improves as model size increases on the GINC dataset

Is the HMMs assumption necessary?

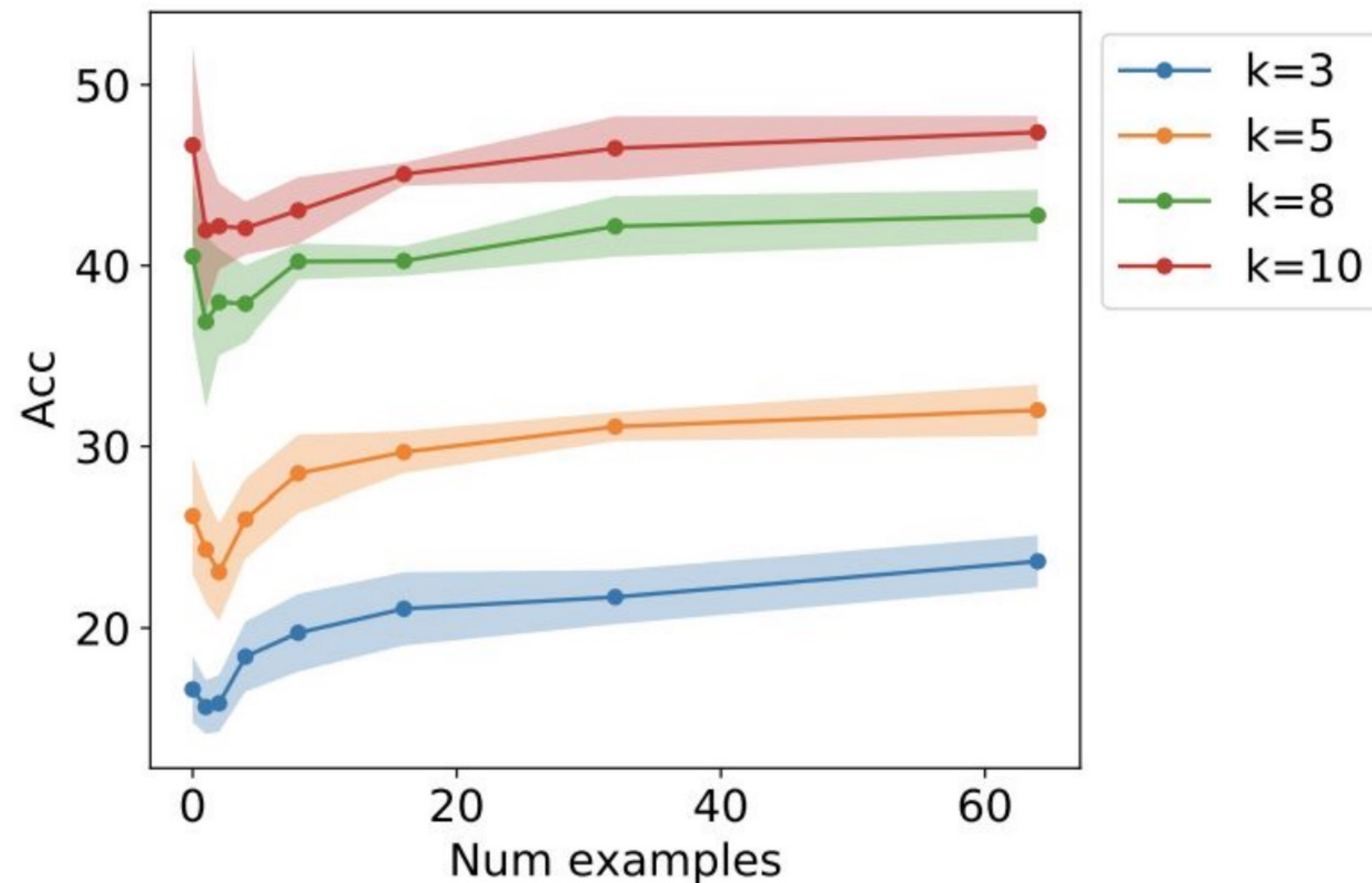


When pre-trained with only one concept, in-context learning fails.



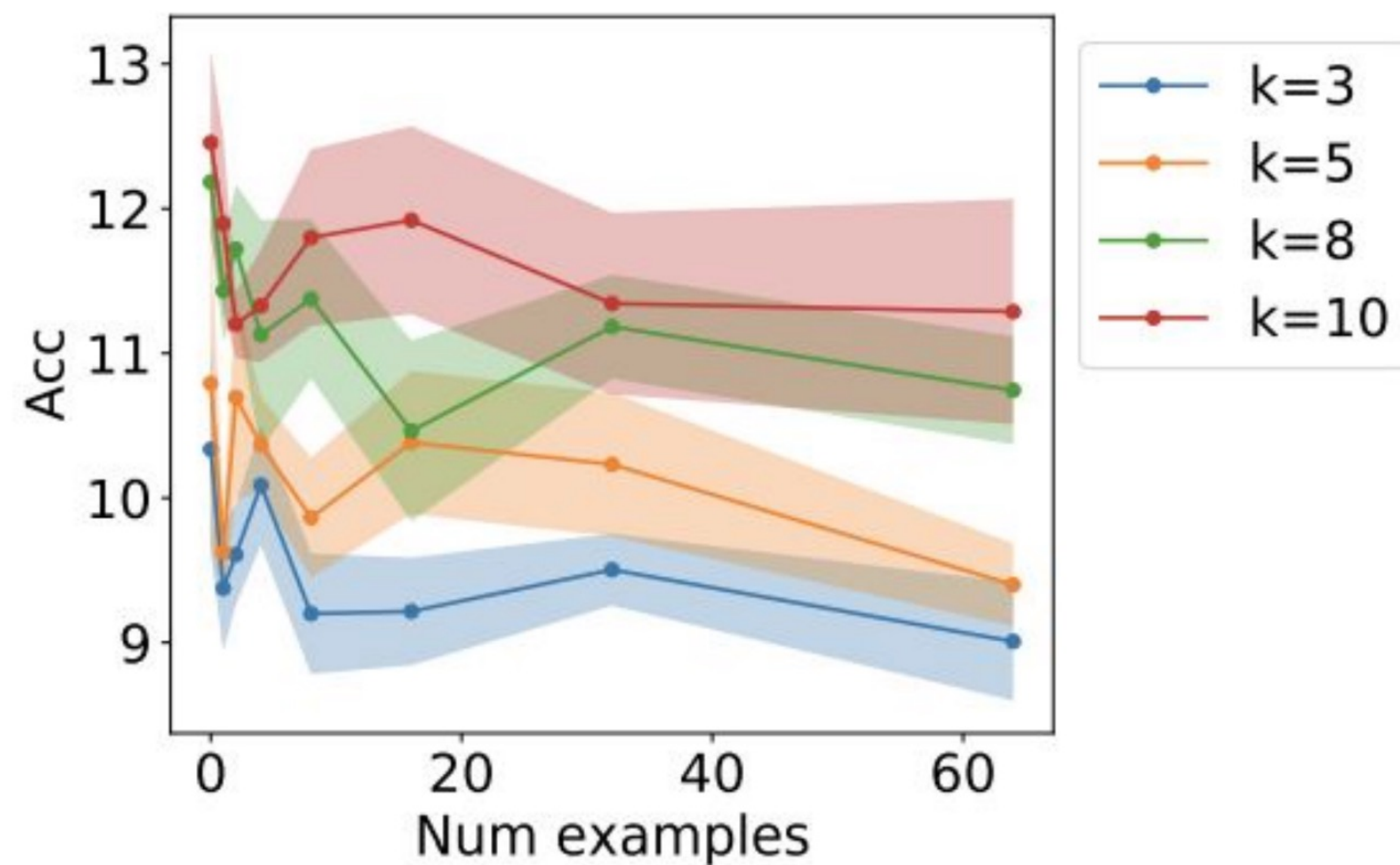
When the pre-training data has random transitions, in-context learning fails.

Zero-shot vs One-shot



- In some settings, few-shot accuracy is initially worse than zero-shot accuracy, but can recover with more examples.
- Mirroring the behavior of GPT-3 on some datasets such as LAMBADA, HellaSwag, PhysicalQA, RACE-m, CoQA/SAT
- Especially because the transition probabilities in GINC are lower entropy

Unseen Concepts



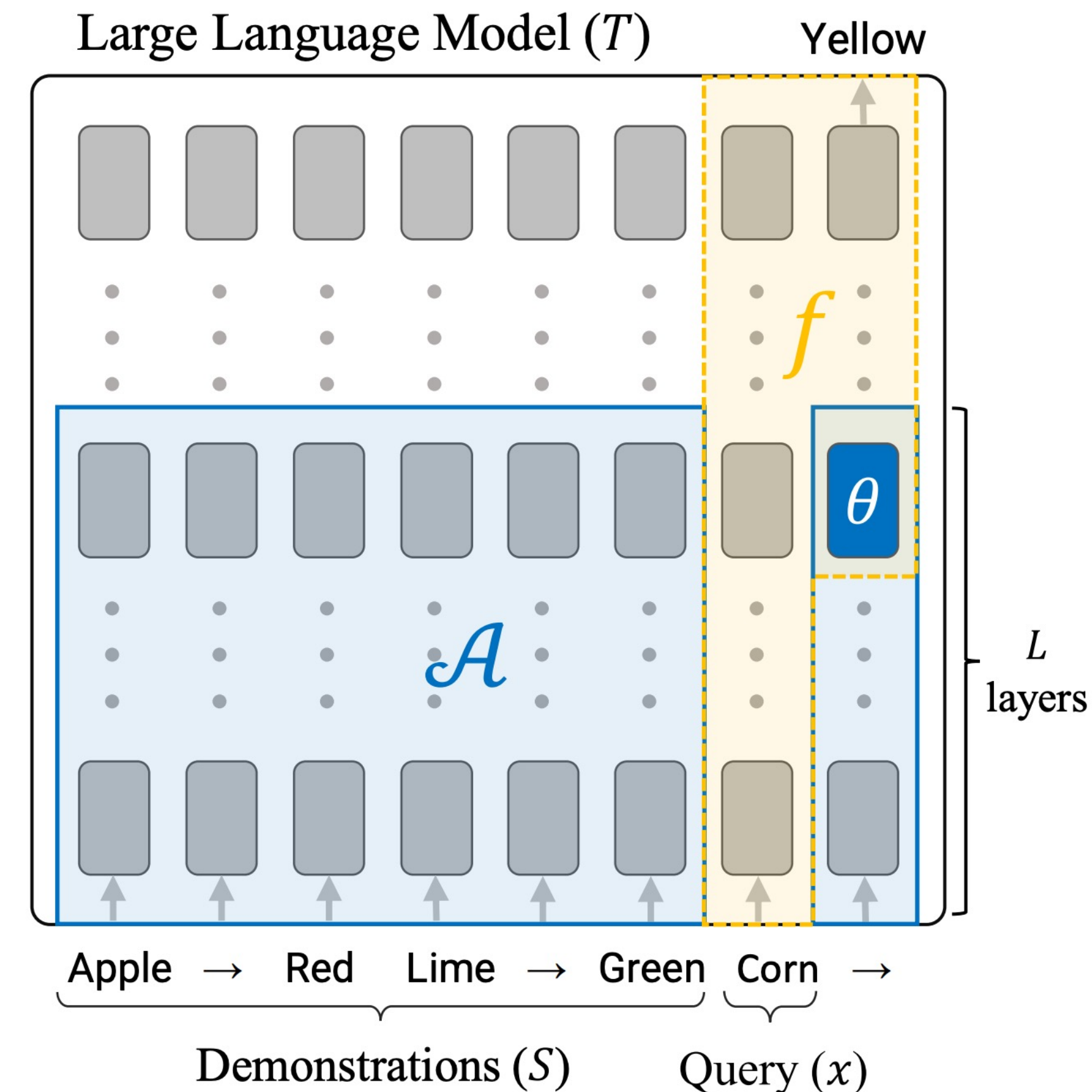
When prompts are from random unseen concepts, in-context learning fails to extrapolate.

Understanding in-context learning

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ICL creates task vectors

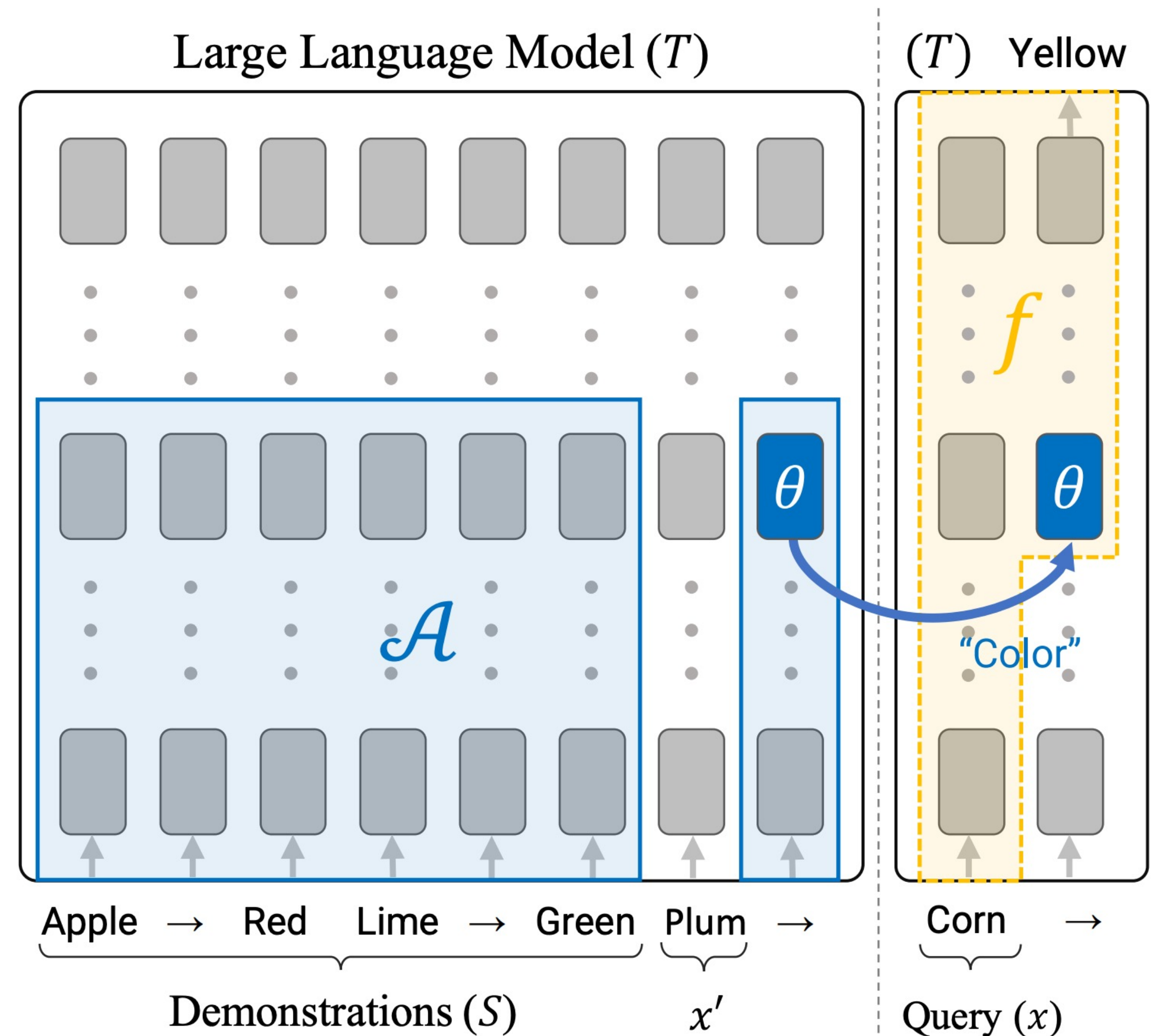
- A decoder-only transformer T is applied on the demonstrations S and the input x
- $T([S, x]) = f(x; A(S))$
 - first maps S into a “task vector” $\theta = A(s)$, independent of x .
 - Then, maps x to the output, based on $\theta \equiv A(S)$, without direct dependence on S .



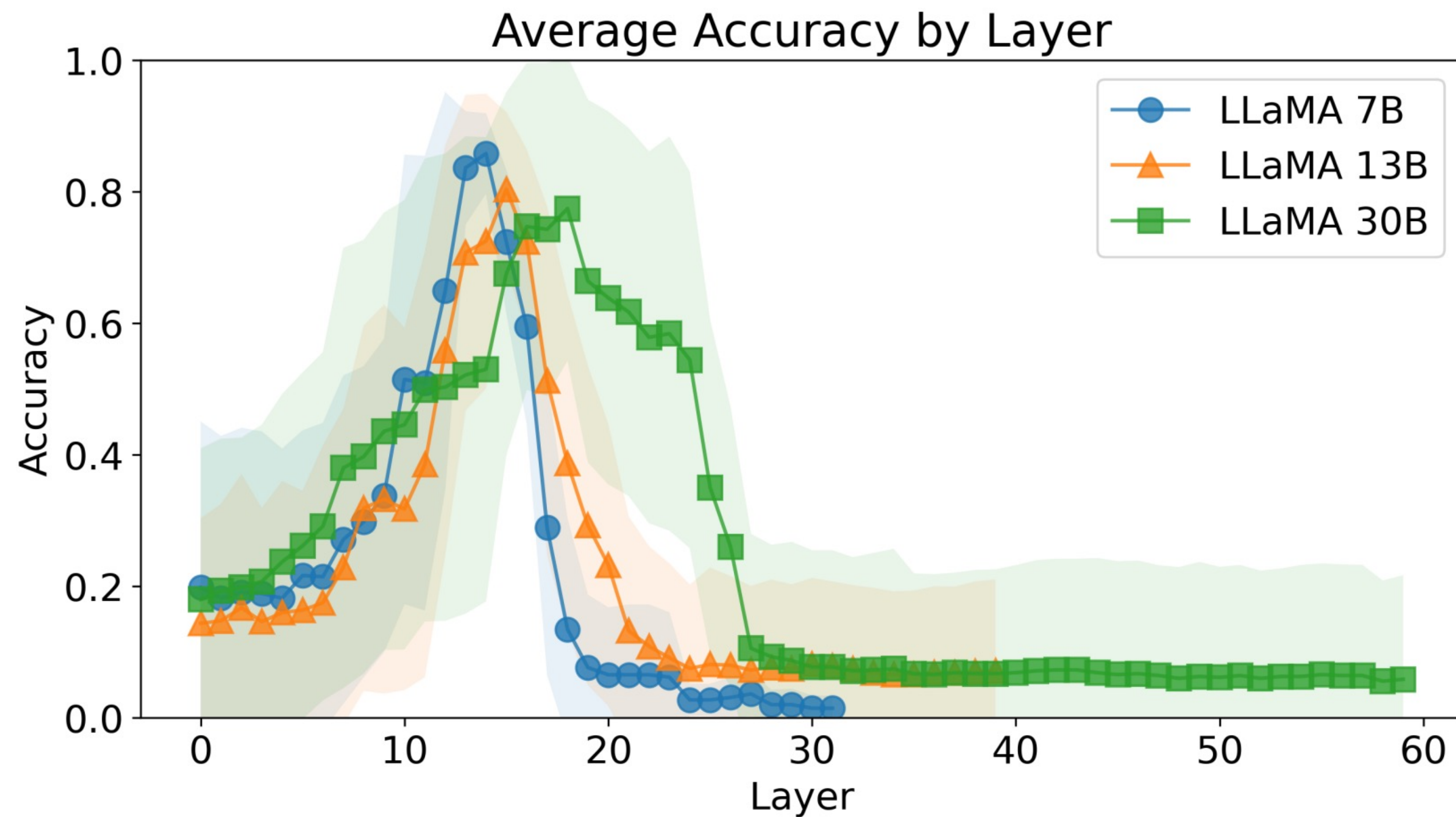
The function f is very similar to the original transformer applied to x without demonstrations but instead modulated by θ

How to separate A and f ?

- Consider the layer L where A ends and f begins
- A generates θ using a dummy x'
- $f(\cdot; \theta)$ is applied to x by running the transformer on $[x, \rightarrow]$ with θ patched at layer L of $\rightarrow$.

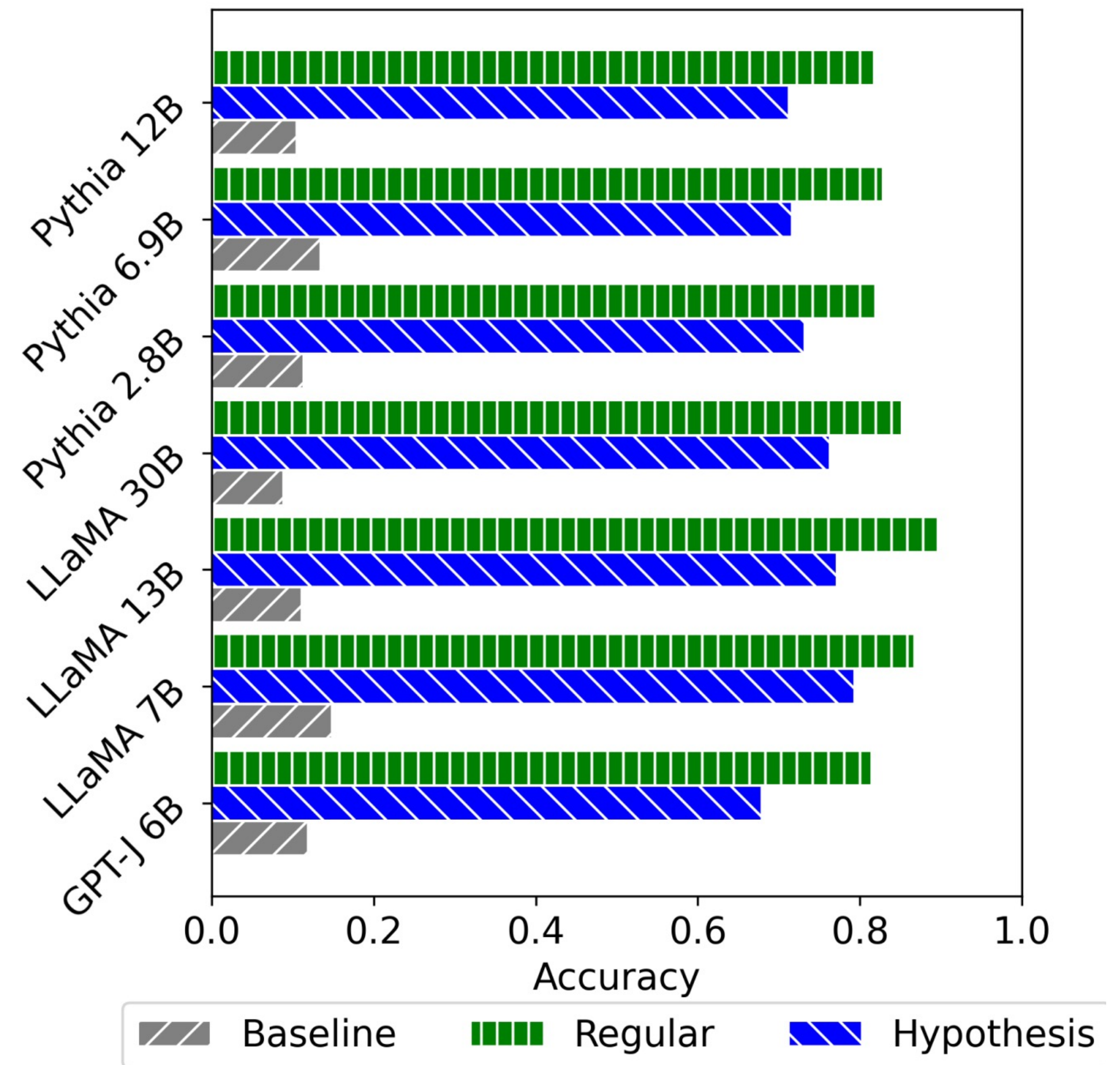


Which layer to extract task vector?



Separation is not such harmful

- Regular ICL: $T([S, x])$
- Hypothesis: A generates θ using a dummy x' , and $f(x; \theta)$ is computed by running the transformer on $[x, \rightarrow]$ with θ patched at layer L of $\rightarrow$.
- Baseline: $T([x, \rightarrow])$ without demonstrations S



Understanding in-context learning

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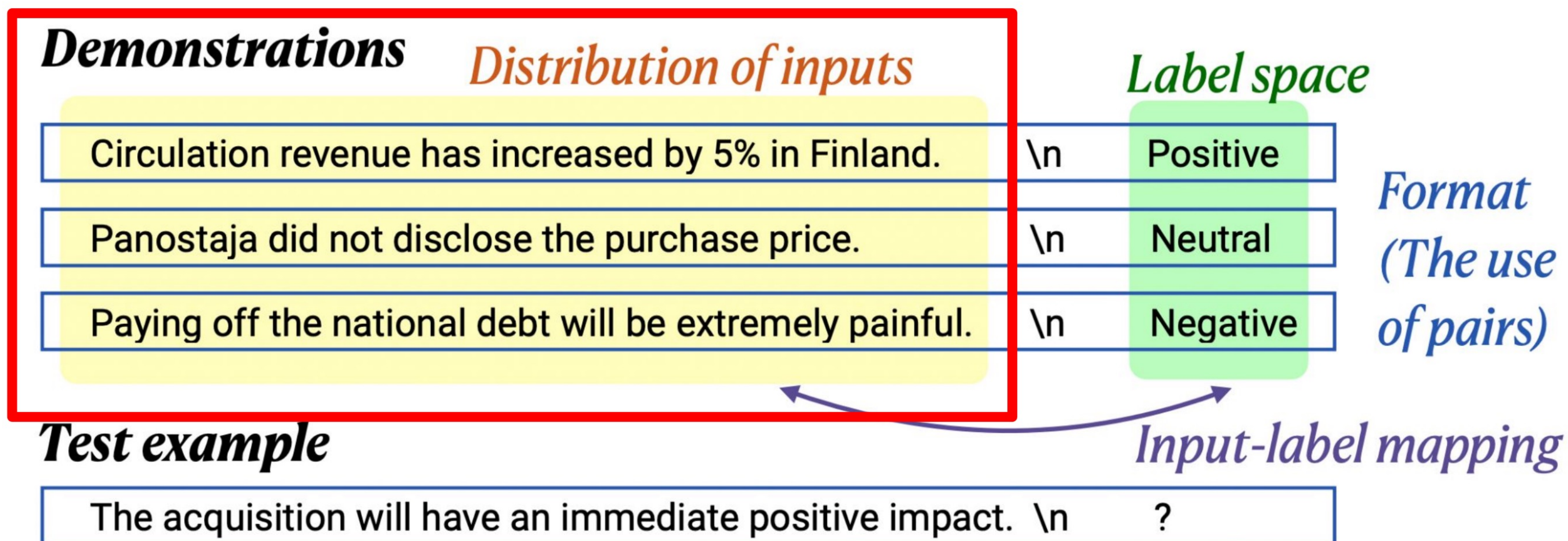
We break the prompt into four parts that provide signal to the model

1 Distribution of Inputs

2 Label Space

3 Input-label Mapping

4 Format



Demonstrations

Distribution of inputs

Circulation revenue has increased by 5% in Finland.	\n	Positive
Panostaja did not disclose the purchase price.	\n	Neutral
Paying off the national debt will be extremely painful.	\n	Negative

Label space

*Format
(The use
of pairs)*

Test example

The acquisition will have an immediate positive impact.	\n	?
---	----	---

Input-label mapping

Demonstrations

Distribution of inputs

Label space

Circulation revenue has increased by 5% in Finland.	\n	Positive
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Input-label mapping

Demonstrations

Distribution of inputs

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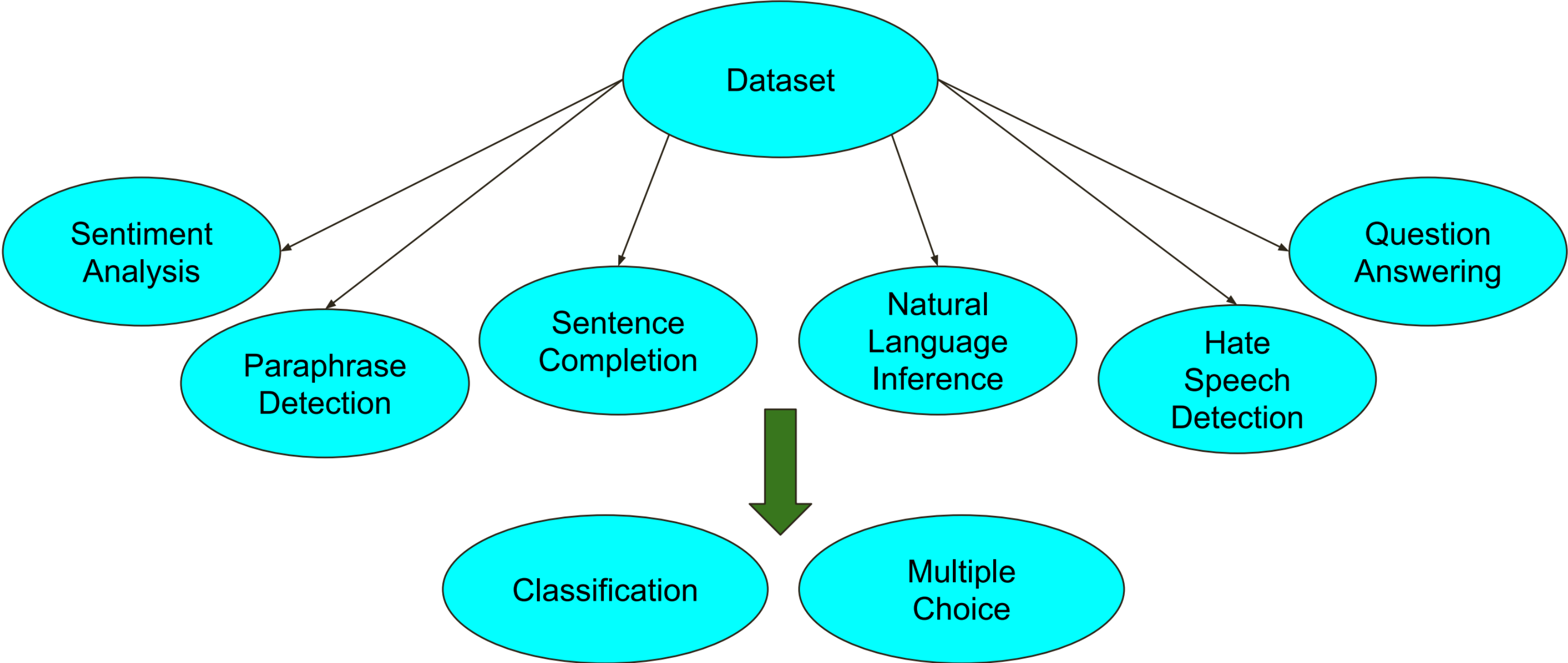
*Format
(The use
of pairs)*

Test example

The acquisition will have an immediate positive impact.	\n	?
---	----	---

Input-label mapping

Datasets



Classification Tasks	
Sentiment Analysis	- financial_phrasebank - poem_sentiment
Paraphrase detection	- medical_questions_pairs - glue-mrpc
Natural language inference	- glue-wnli - climate fever
Question Answering	- quarel - openbookqa

Multiple Choice Tasks	
Hate speech detection	- hate_speech18 - ethos-national_origin
Sentence completion	- codah - superglue-copa

Evaluation Methodology

- Metrics
 - Classification: **Macro-F1**
 - Multiple Choice: **Accuracy**
- Compute per-dataset average across seeds, and report **macro-average over datasets**

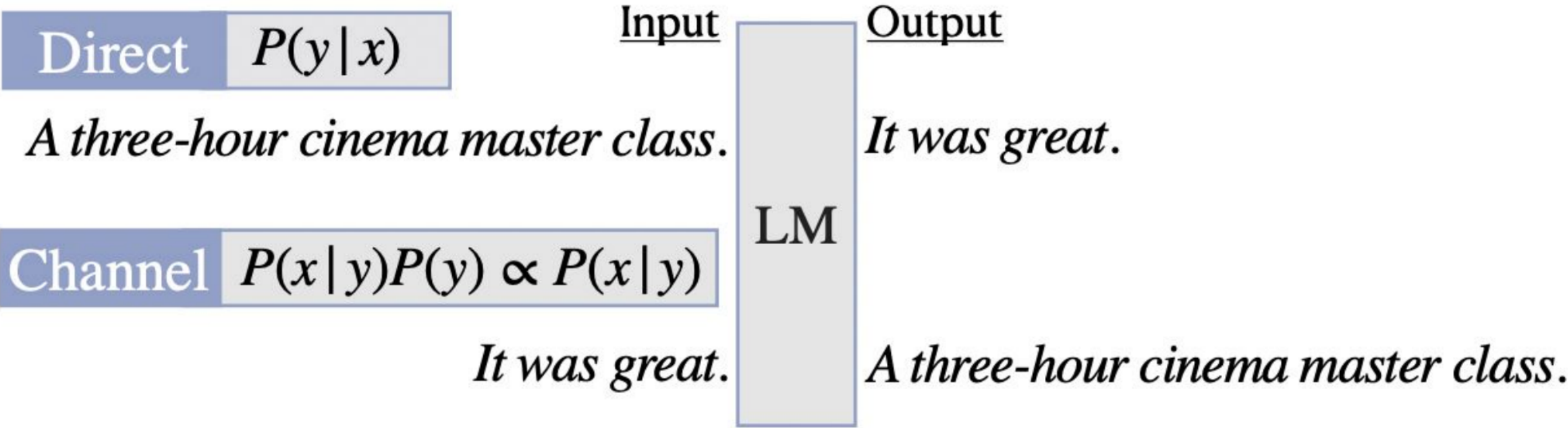


Models

Model	# Params	Public	Meta-trained
GPT-2 Large	774M	✓	✗
MetaICL	774M	✓	✓
GPT-J	6B	✓	✗
fairseq 6.7B [†]	6.7B	✓	✗
fairseq 13B [†]	13B	✓	✗
GPT-3	175B [‡]	✗	✗

Direct vs. Channel Models

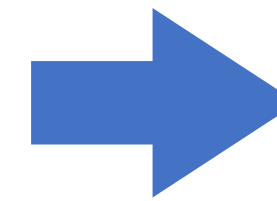
$(x, y) = (\text{"A three-hour cinema master class."}, \text{"It was great."})$



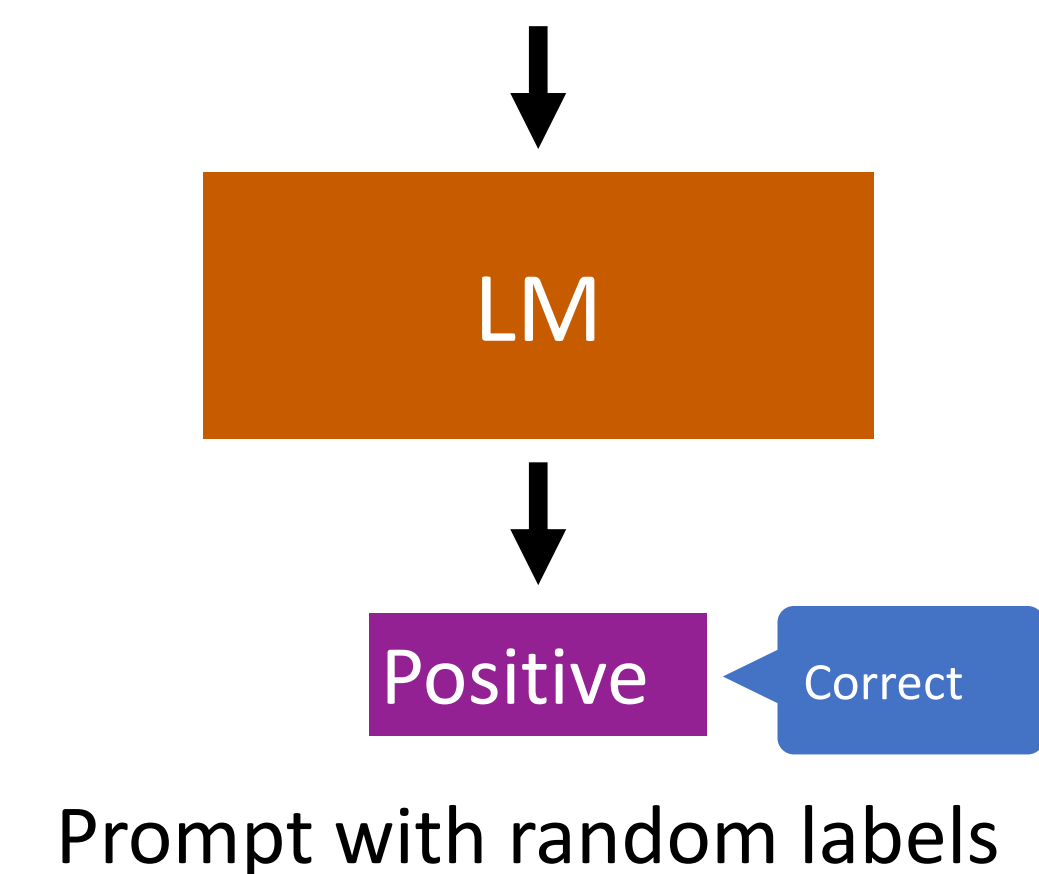
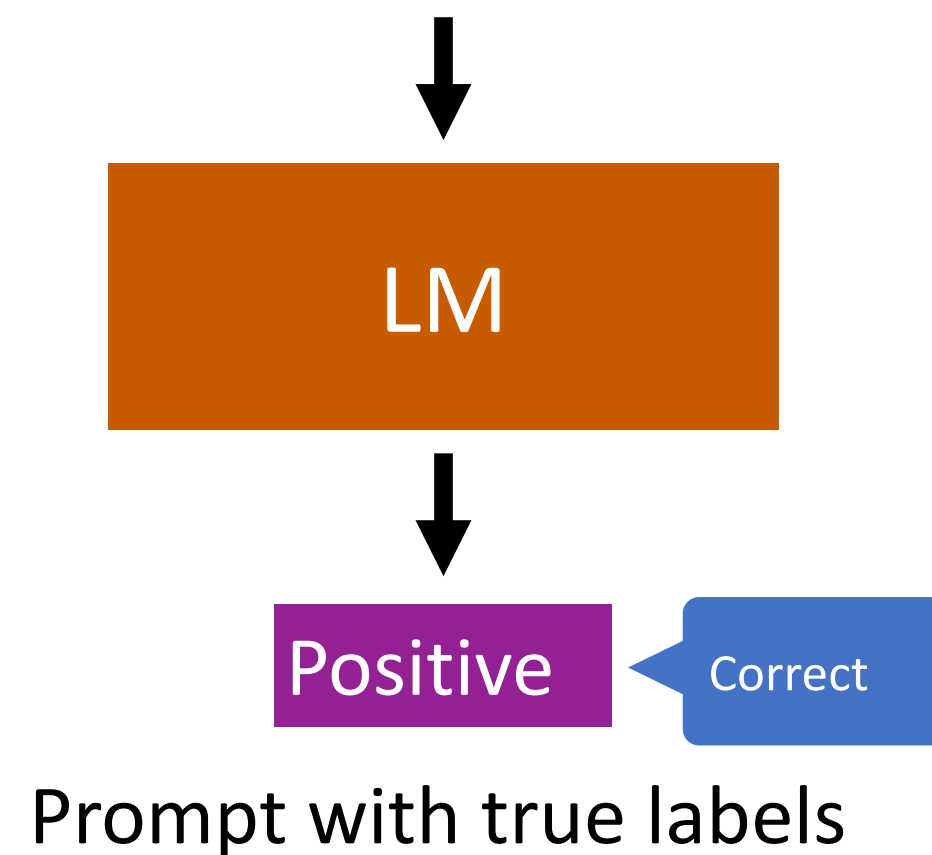
$P(y|x) \longrightarrow P(x|y)$

True Labels vs Random Labels

Circulation revenue has increased by 5% in Finland.	Positive
Panostaja did not disclose the purchase price.	Neutral
Paying off the national debt will be extremely painful.	Negative
The acquisition will have an immediate positive impact.	??



Circulation revenue has increased by 5% in Finland.	Neutral
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Paying off the national debt will be extremely painful.	Positive
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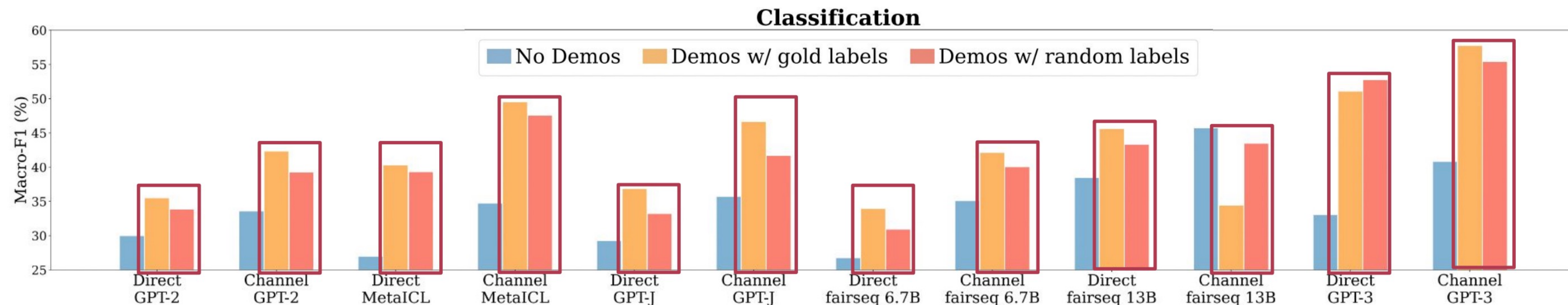


Note

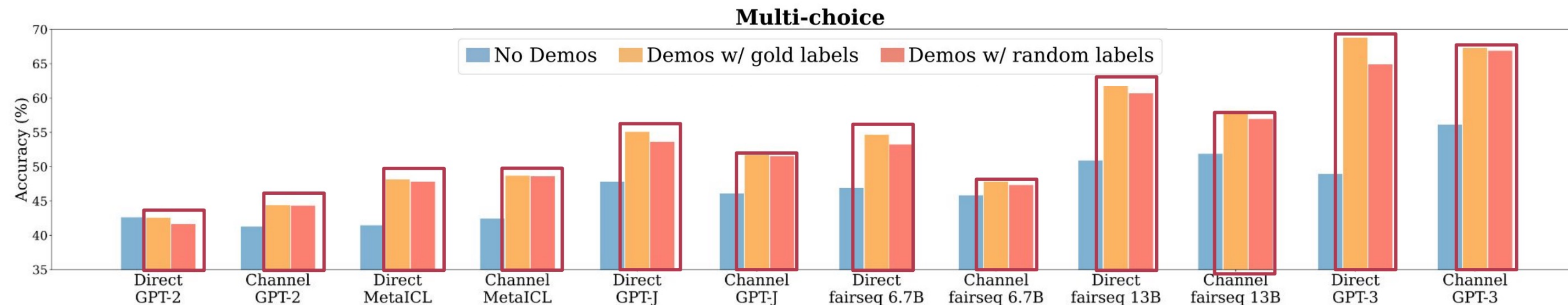
1. Randomly sample a label from the correct label space
2. Assign the label to the example

Results

concept ← در کمال تعجب دقت خیلی افت نکرده



Comparisons between no-examples (blue), examples with ground truth outputs (yellow) and examples with random outputs (red)



Comparisons between no-examples (blue), examples with ground truth outputs (yellow) and examples with random outputs (red)

Models see small performance drop in the range of 0–5% absolute with random labels

Results Takeaways

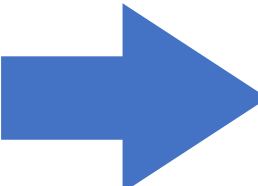
- Ground truth input-label mapping in the prompt is not as important as we thought
- Model is not recovering the expected input-label correspondence for the task from the input-label pairings

Question

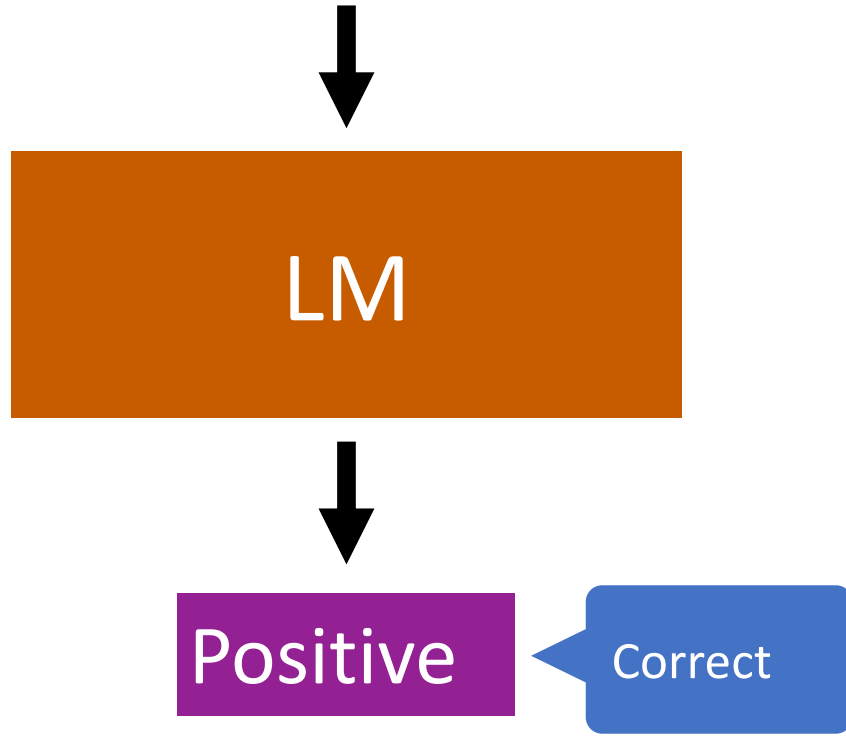
Is this result consistent in other setups?

Does the number of correct labels matter?

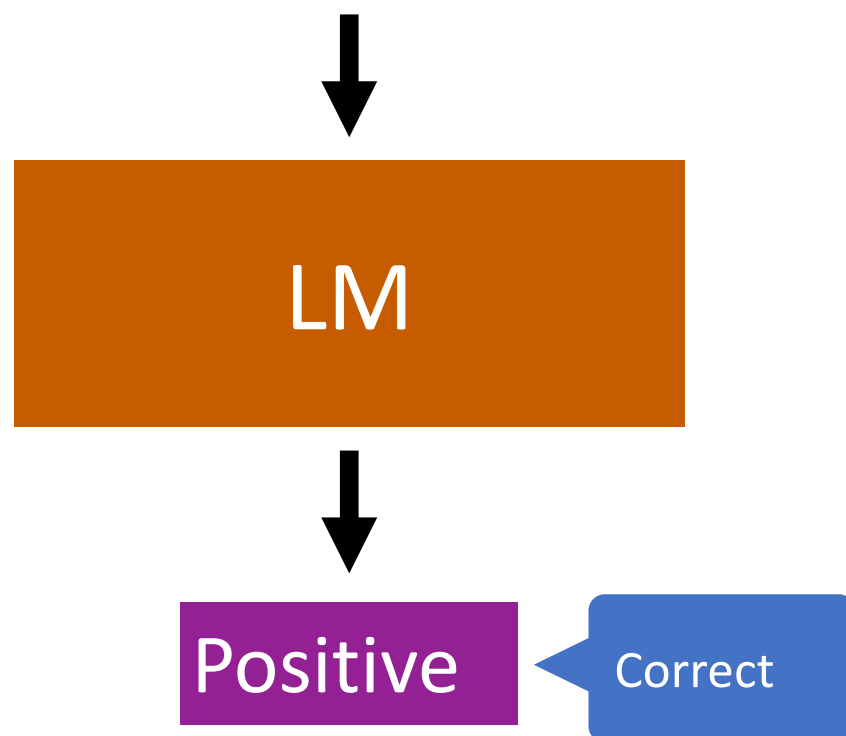
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Prompt with true labels

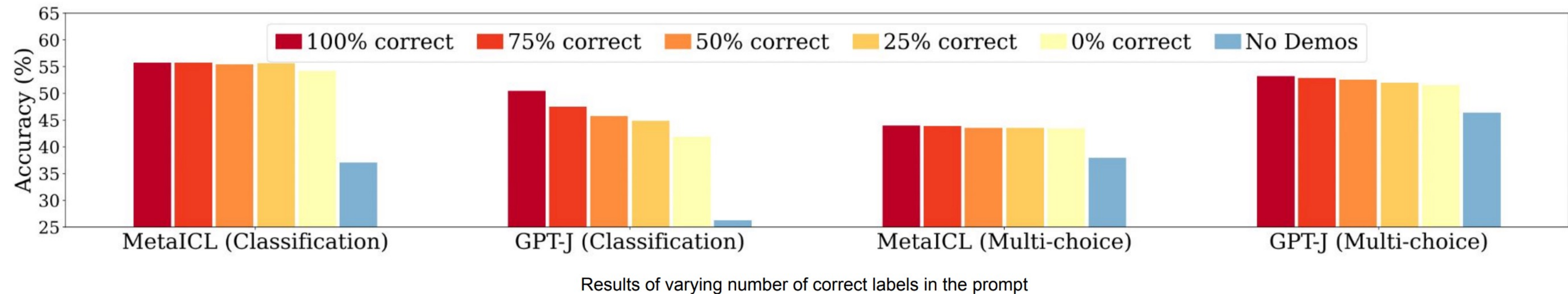


Prompt with random labels

Note

1. Vary the number of correct labels in examples

Results

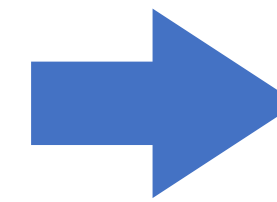


Model performance is fairly insensitive to the number of correct labels

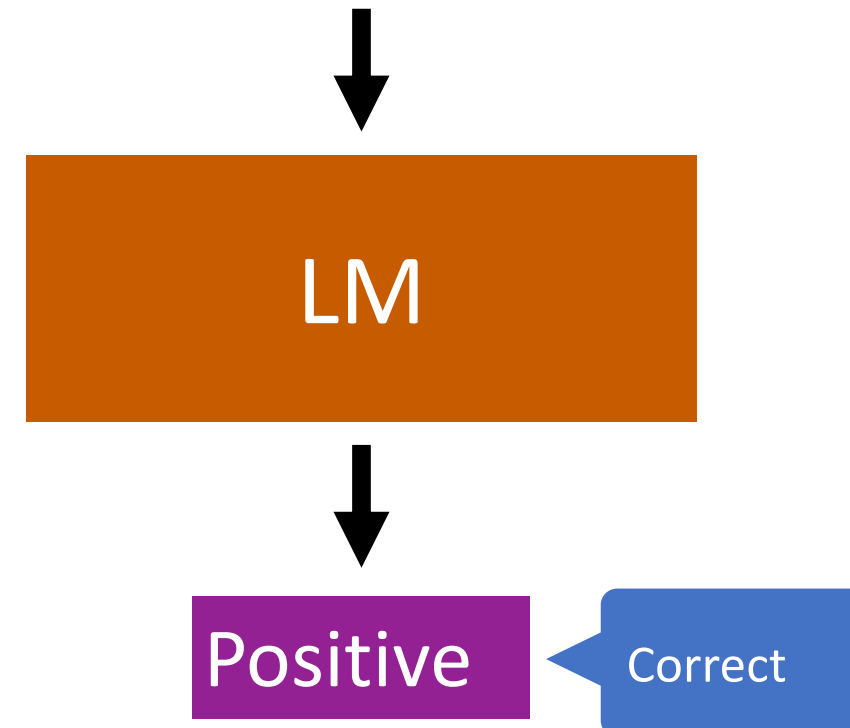
Using incorrect labels is better than no examples

Varying the Number of Examples

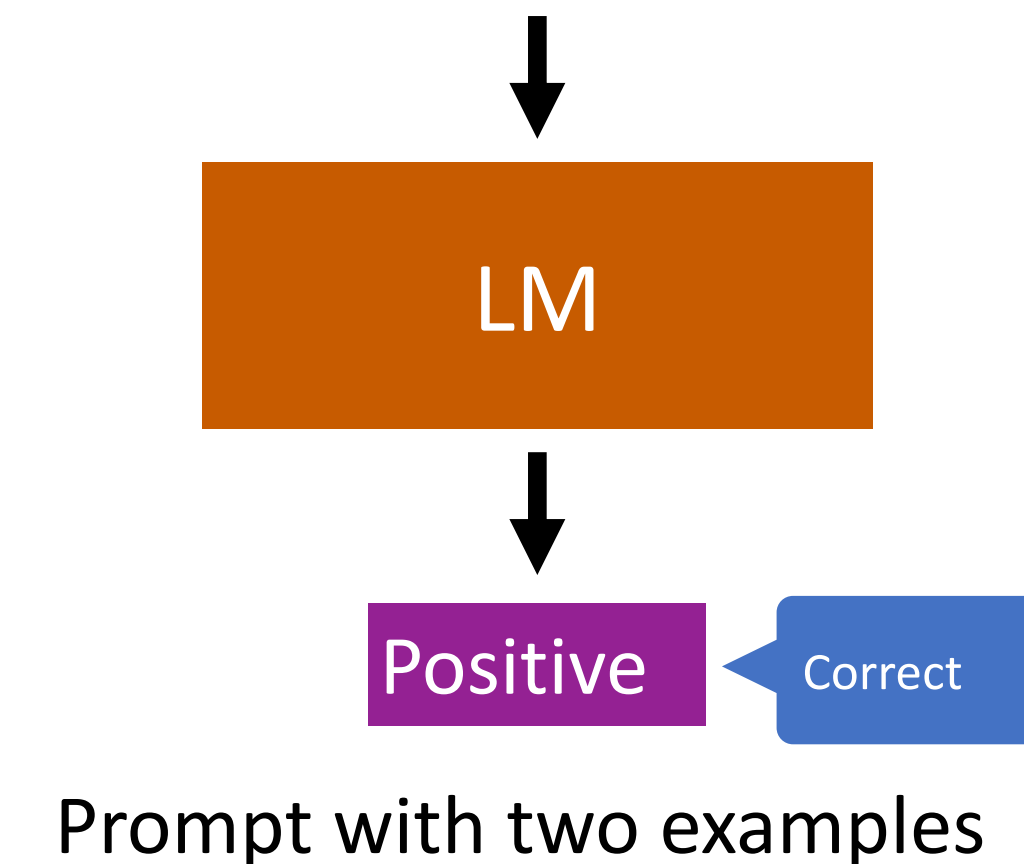
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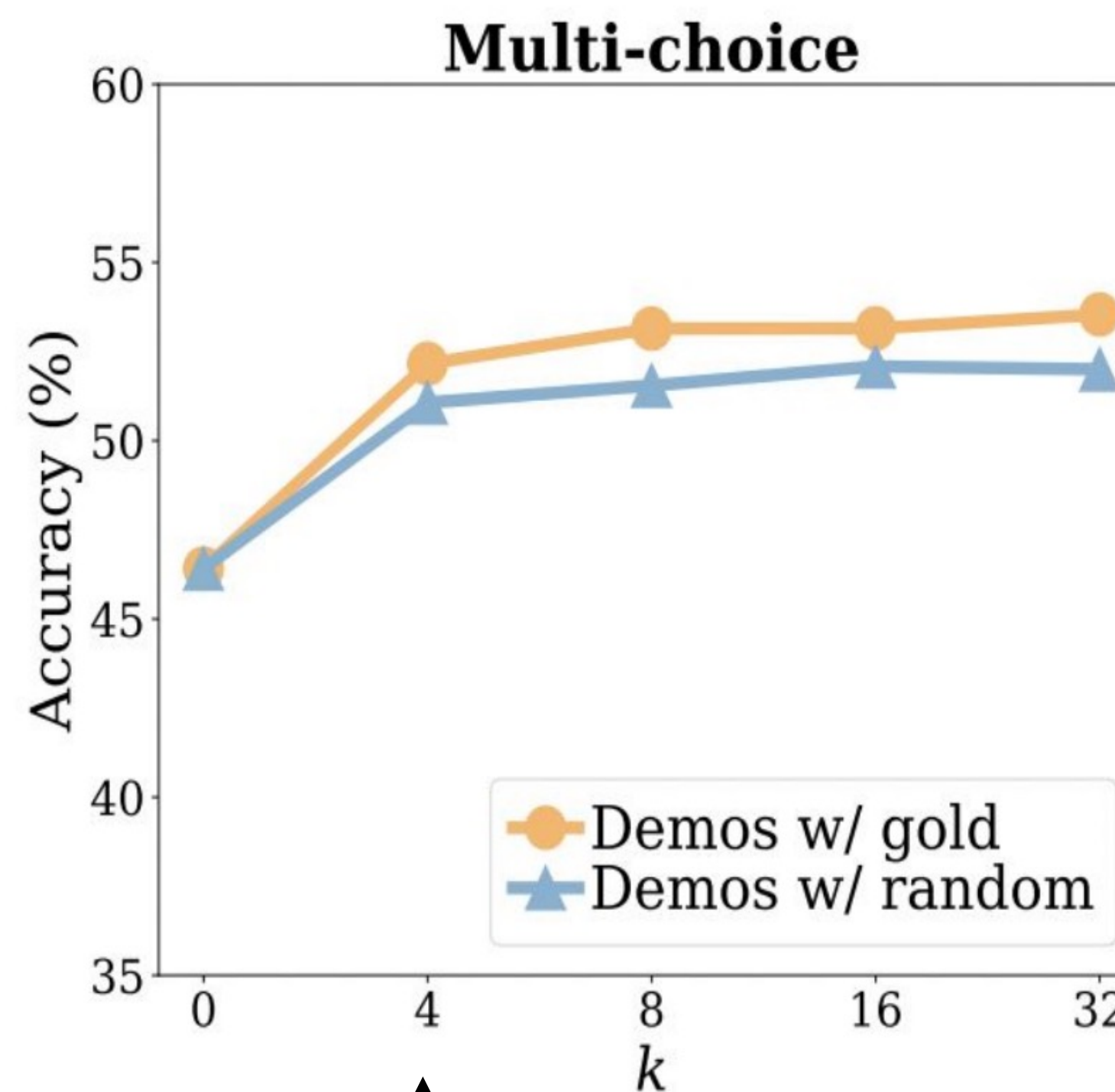
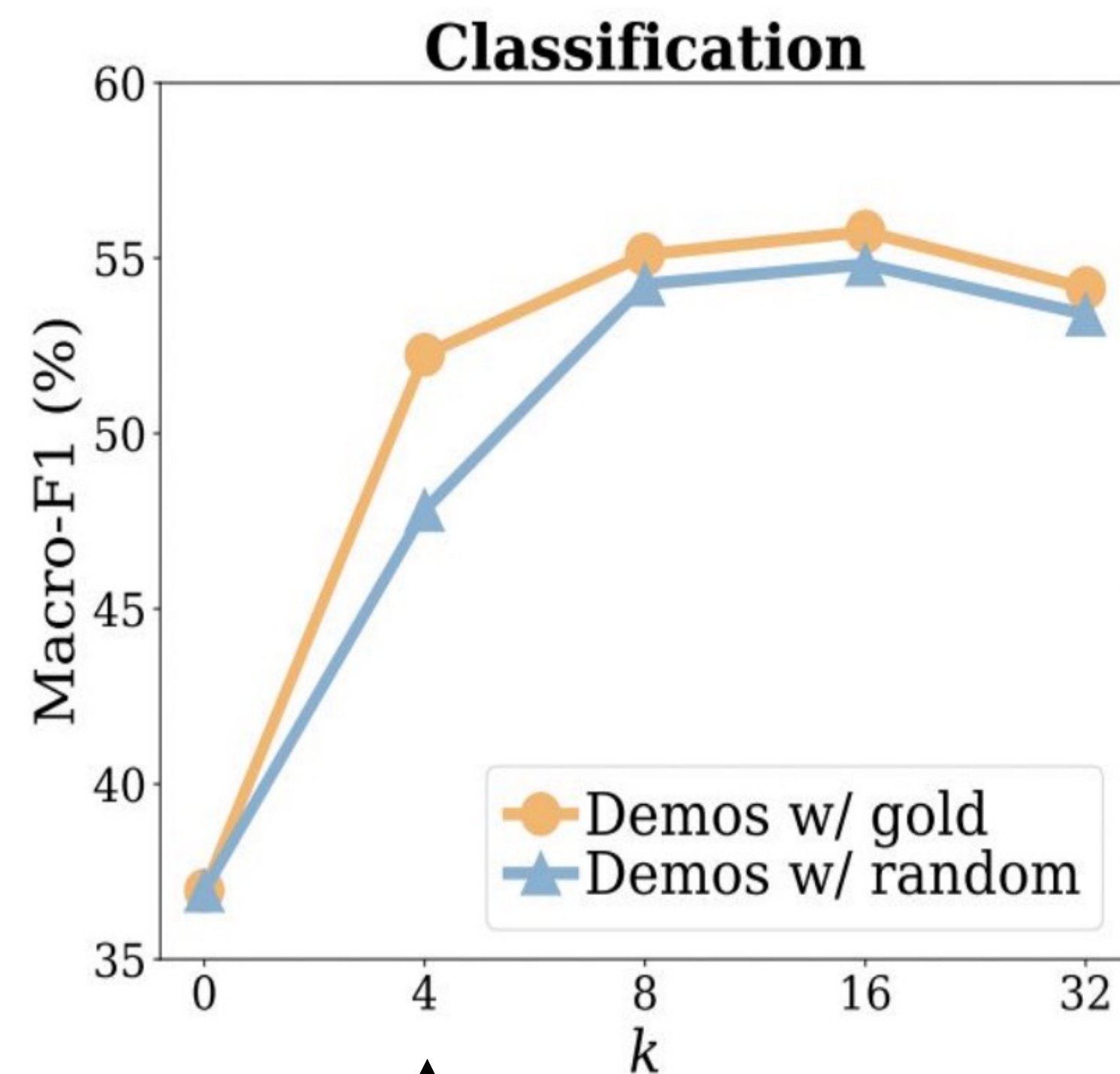
Prompt with three examples



Note

Measure whether the results of using random labels is consistent across differing number of examples

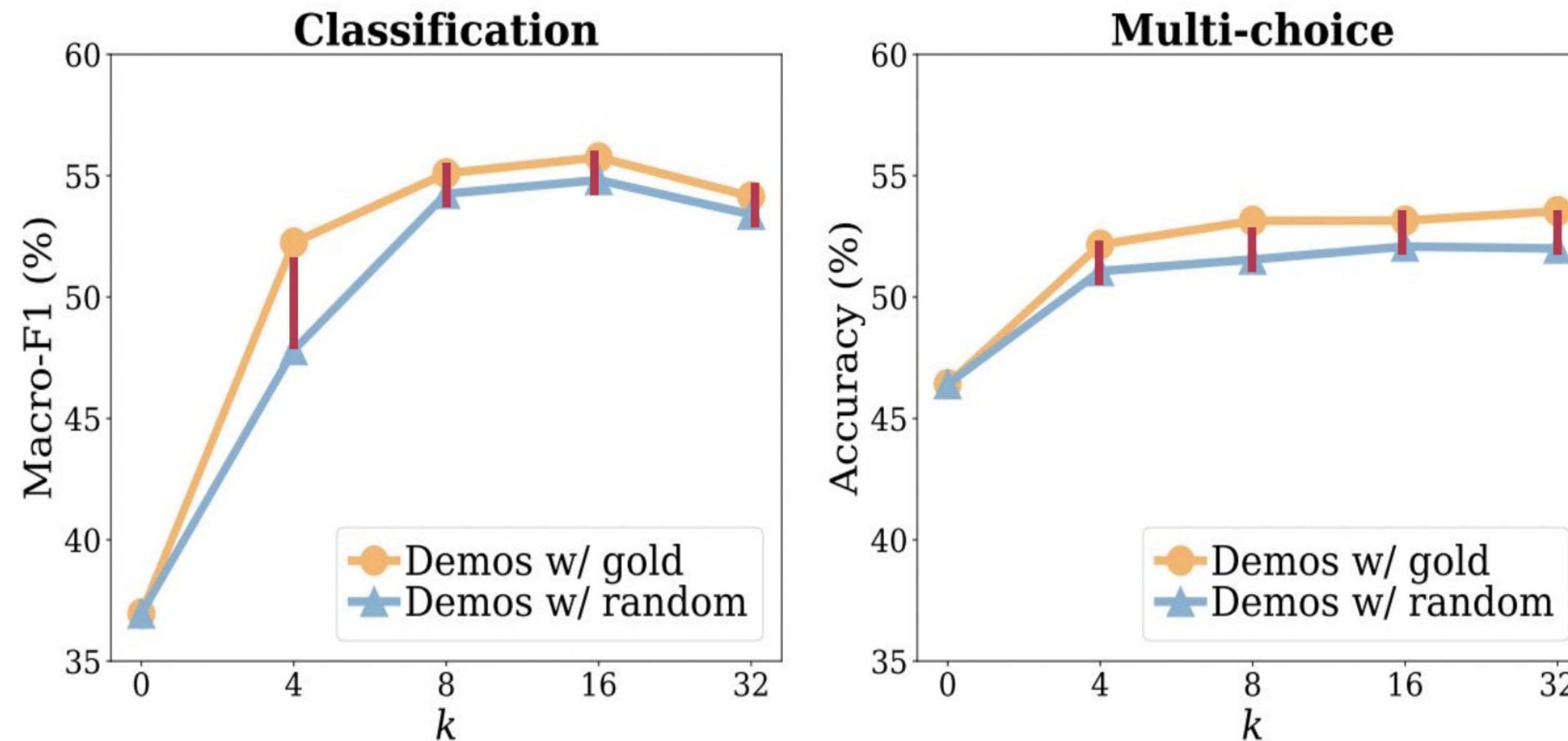
Results



Ablations on varying numbers of examples (k) in the prompt

Using **small number** of examples with **random labels** is better than no **examples**

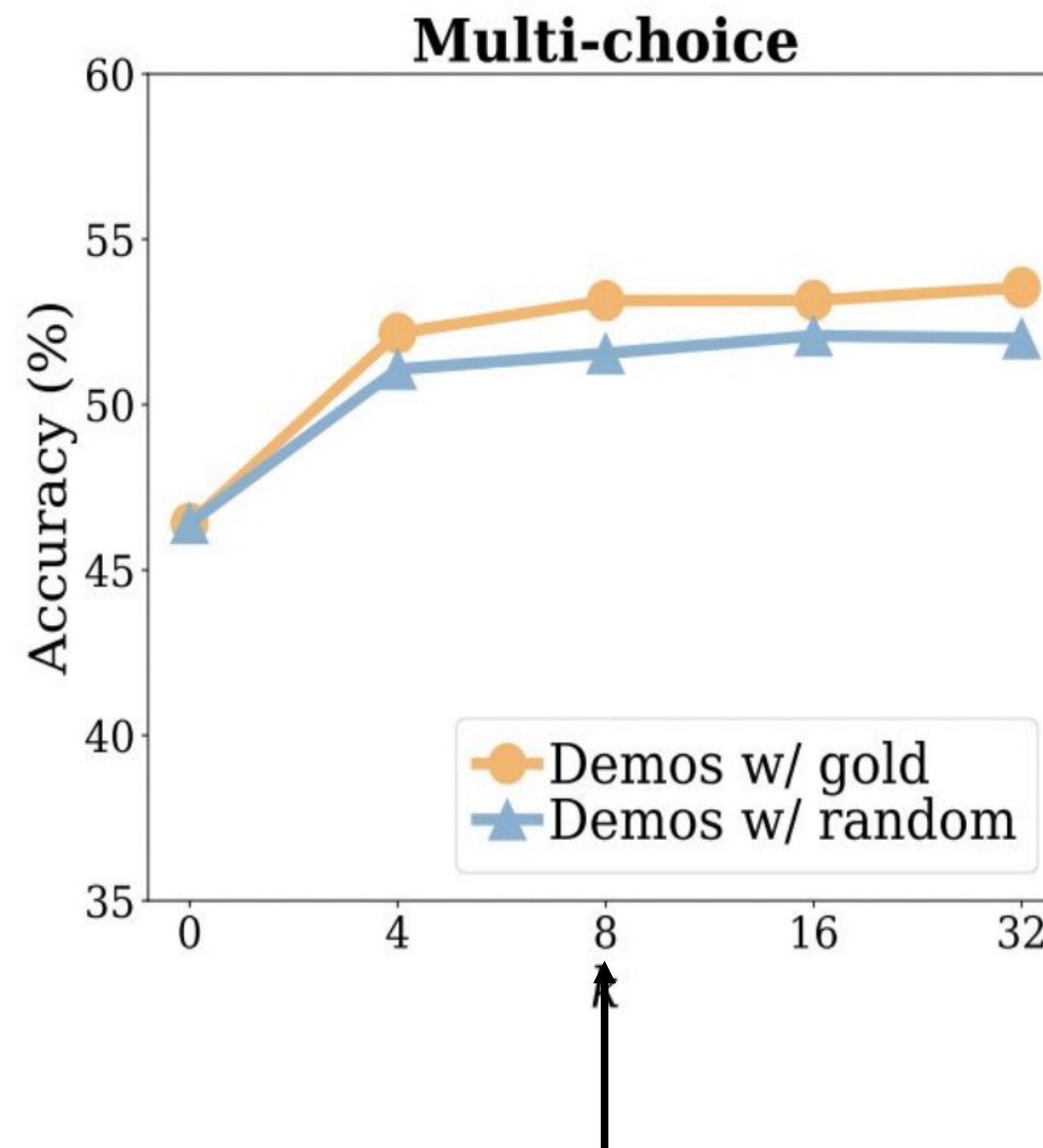
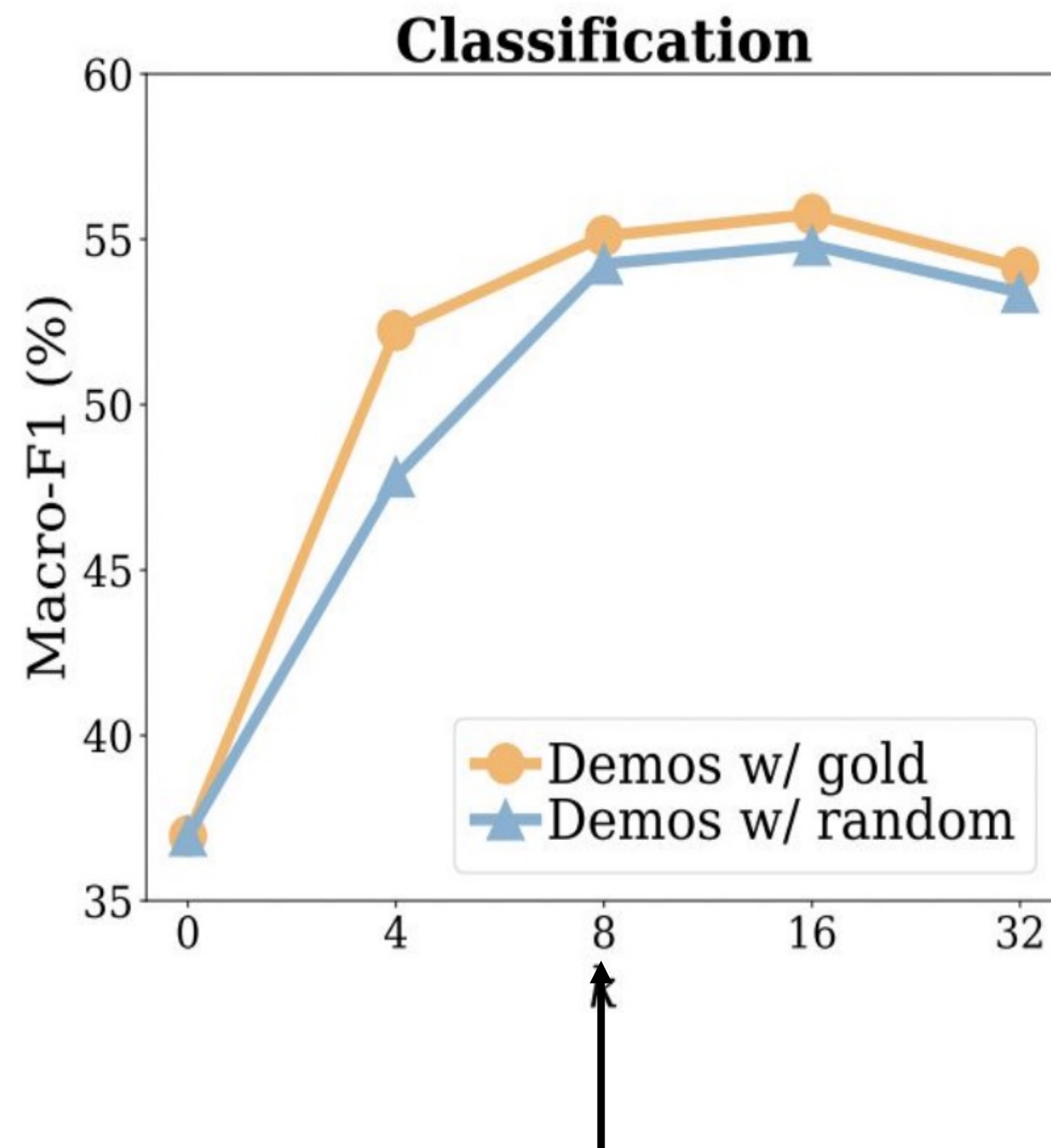
Results



Ablations on varying numbers of examples (k) in the prompt

Performance drop from using gold labels to using random labels is **consistently small** across varying number of examples (k), ranging from 0.8–1.6%

Results



Ablations on varying numbers of examples (k) in the prompt

More examples even with random labels improves model performance except beyond a threshold

Model performance does not increase much as k increases when $k \geq 8$ even for gold labels

Using Better Templates

Dataset	Type	Example
Tweet_eval-hate	Minimal	The Truth about #Immigration \n {hate non-hate}
	Manual	Tweet: The Truth about #Immigration \n Sentiment: {against favor}

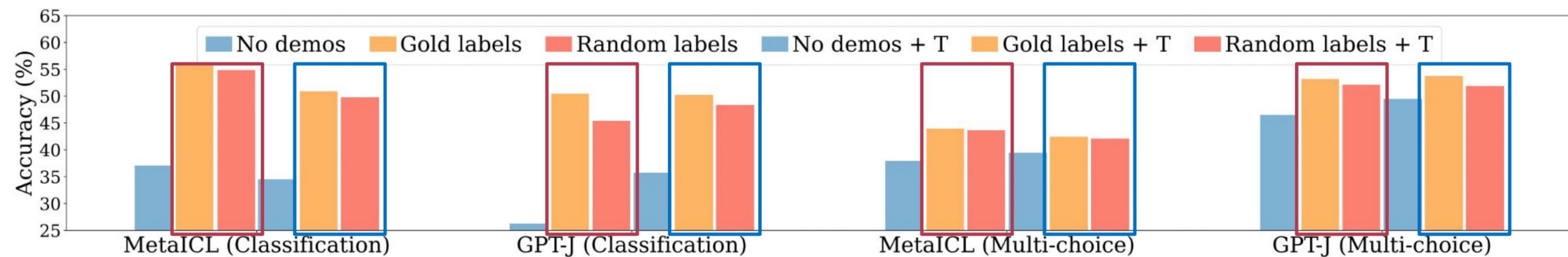
Example of minimal and manual templates

- **Minimal** templates follow a conversion procedure (dataset-agnostic)
- **Manual** templates are written in a dataset-specific manner

Note

Measure whether the results of using random labels is consistent when using manual templates

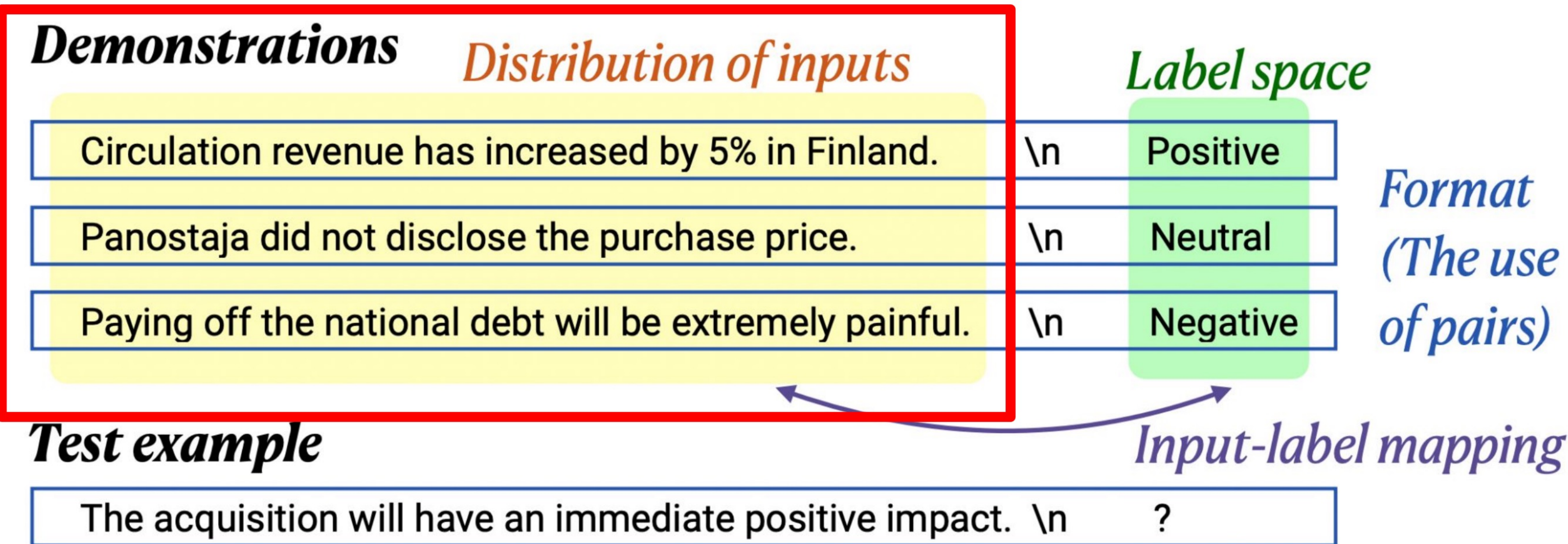
Results



Results with minimal templates and manual templates. '+T' indicates that manual templates are used.

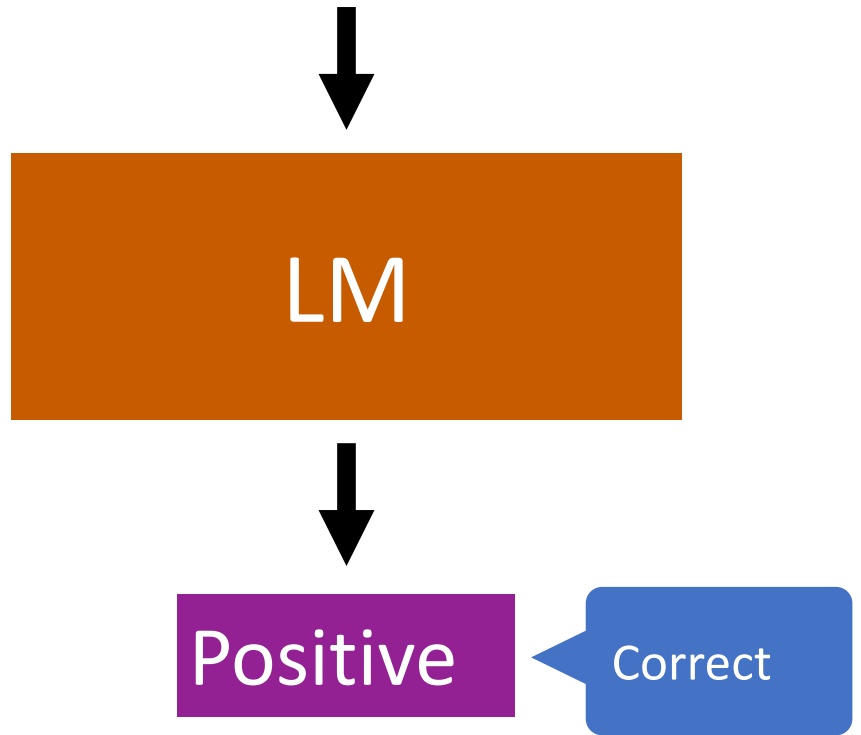
Random labels still minimally hurt performance with manual templates

Distribution of Inputs

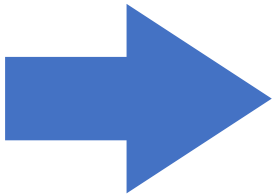


Using out-of-distribution input text

Circulation revenue has increased by 5% in Finland.	Positive
Panostaja did not disclose the purchase price.	Neutral
Paying off the national debt will be extremely painful.	Negative
The company anticipated its operating profit to improve.	??

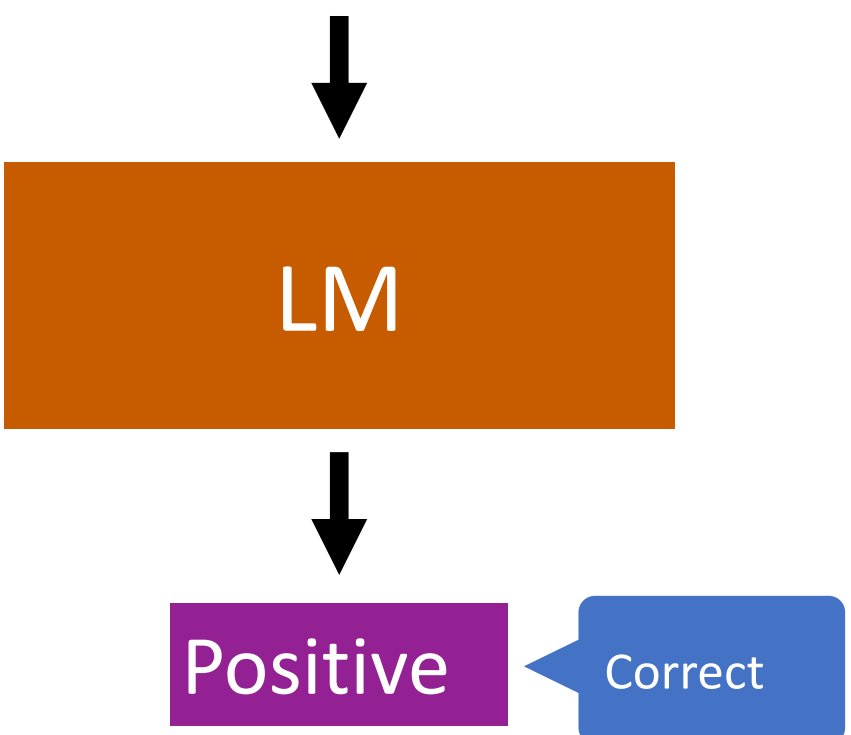


Prompt with in-distribution sentences



Colour-printed lithograph. Very good condition.	Neutral
Many accompanying marketing ... meaning.	Negative
In case you are interested in learning more about ...	Positive
The company anticipated its operating profit to improve.	??

*Randomly Sampled from CC News

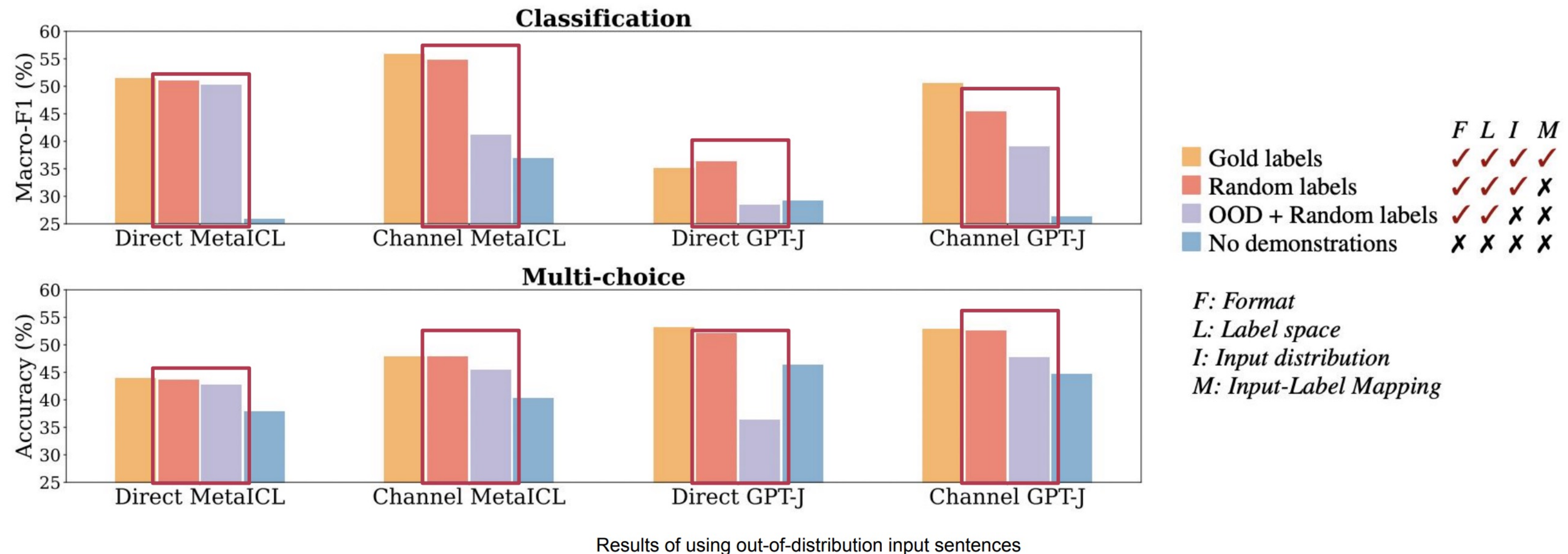


Prompt with out-of-distribution sentences

Note

Input sentences are randomly sampled from an external corpus, replacing the input from the downstream task training data

Seeing in-distribution inputs improves performance



Random sentences result in performance **decreases of up to 16% absolute** compared to using inputs from training data

Label Space

Demonstrations

Distribution of inputs

Label space

Circulation revenue has increased by 5% in Finland.	\n	Positive
Panostaja did not disclose the purchase price.	\n	Neutral
Paying off the national debt will be extremely painful.	\n	Negative

*Format
(The use
of pairs)*

Test example

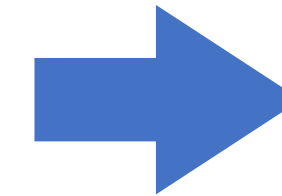
The acquisition will have an immediate positive impact.	\n	?
---	----	---

Input-label mapping

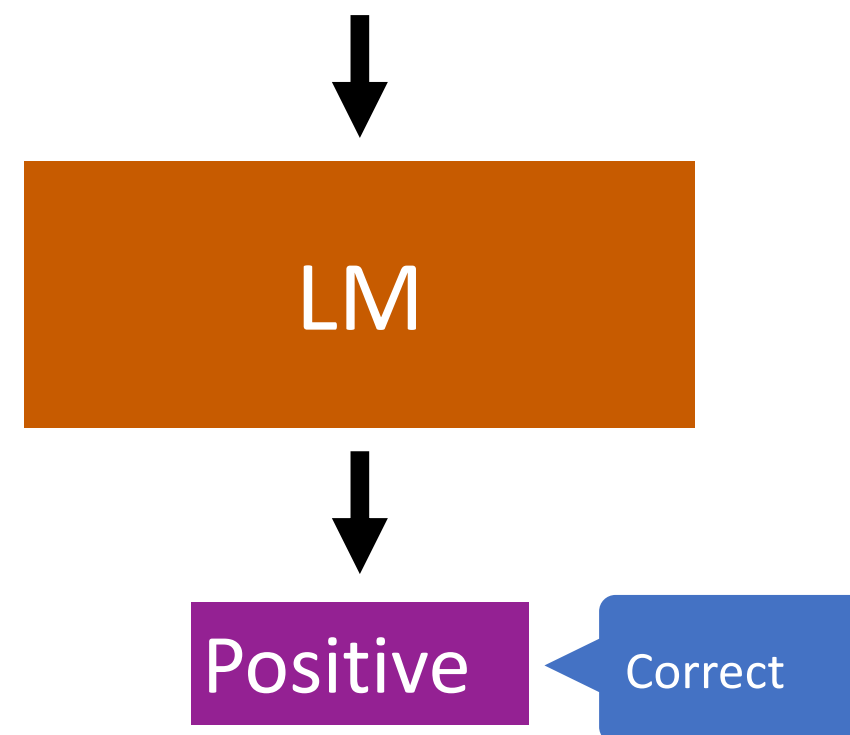
Evaluate the importance of the label space

Using random labels from an incorrect label space

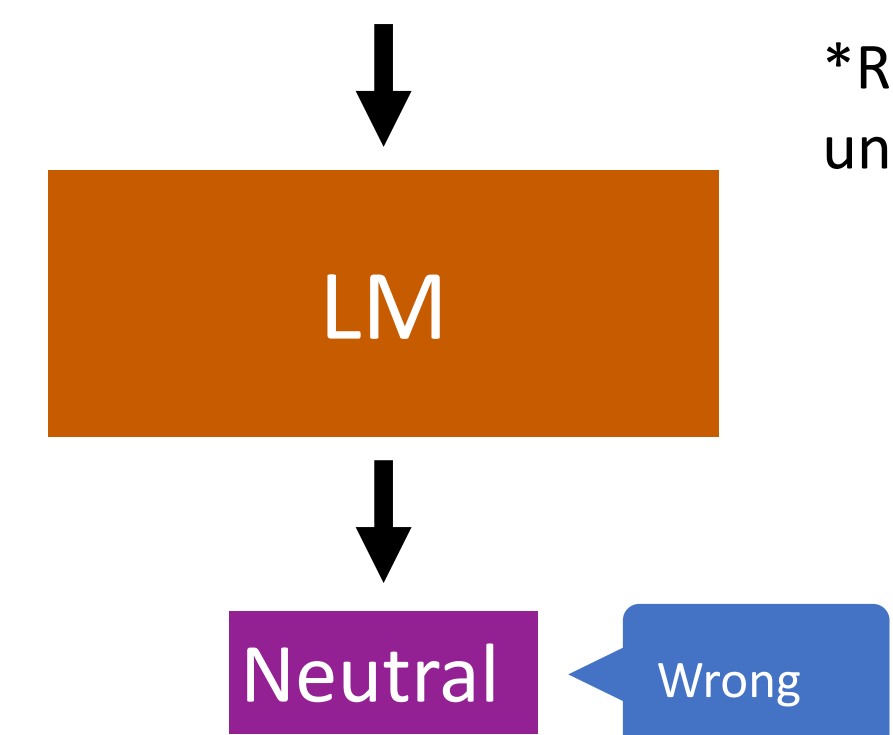
Circulation revenue has increased by 5% in Finland.	Positive
Panostaja did not disclose the purchase price.	Neutral
Paying off the national debt will be extremely painful.	Negative
The company anticipated its operating profit to improve.	??



Colour-printed lithograph. Very good condition.	Unanimity
Many accompanying marketing ... meaning.	Wave
In case you are interested in learning more about ...	Guana
The company anticipated its operating profit to improve.	??



Prompt with true labels



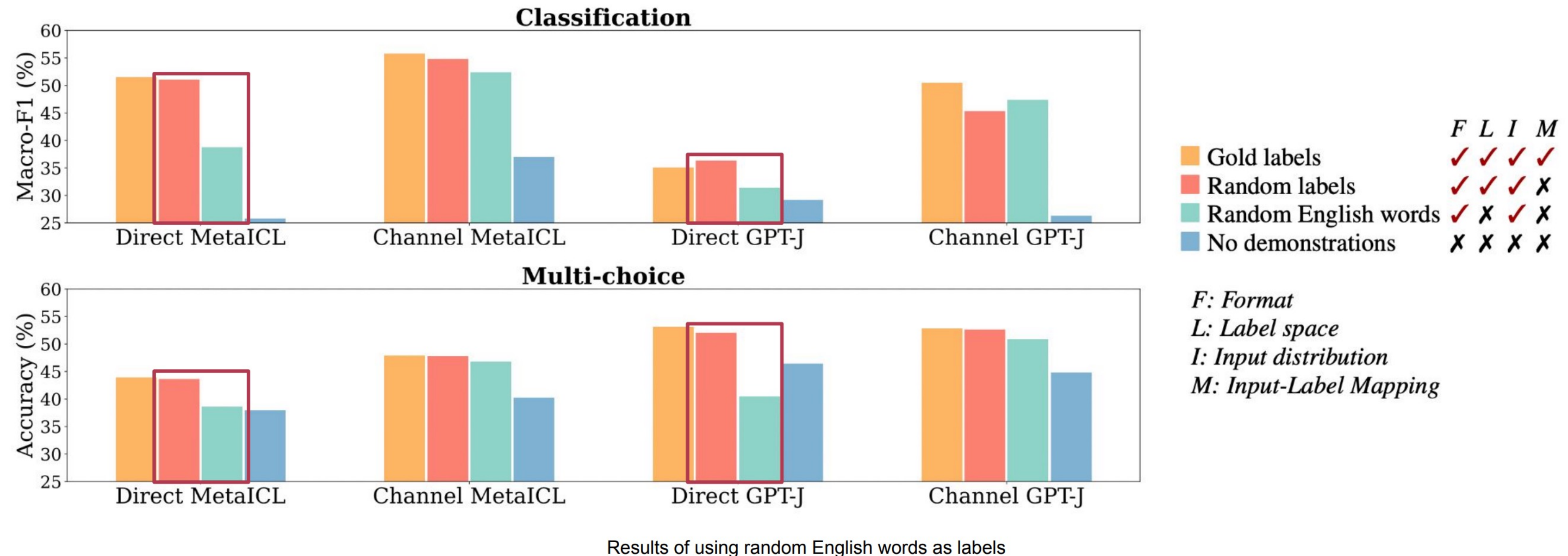
*Random English unigram

Prompt with random English words as labels

Note

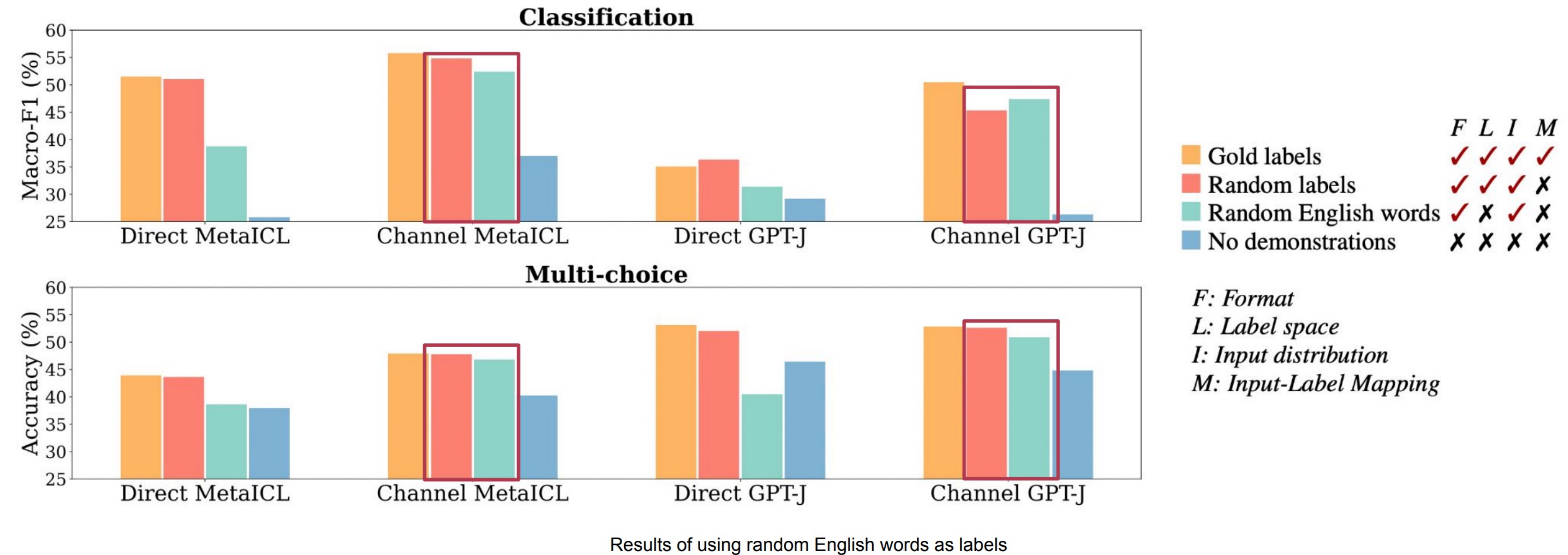
1. Sample a random subset of English words with same size as set of truth labels
2. Labels are replaced with words randomly drawn from this subset

Seeing correct label space is important



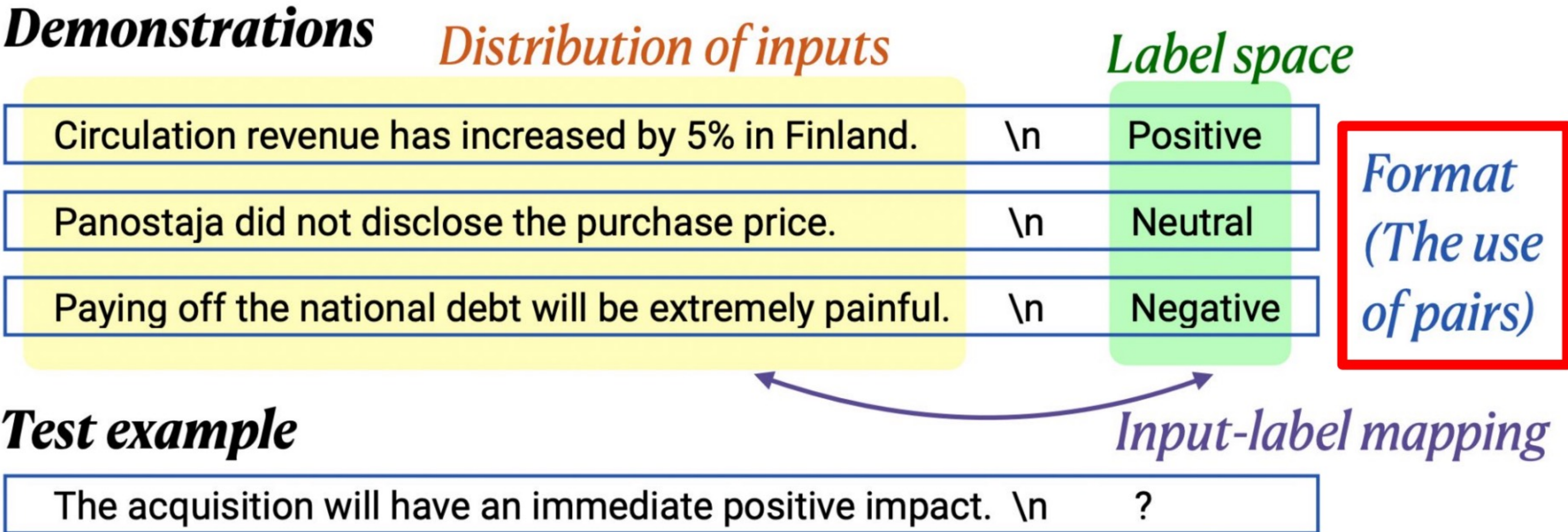
Labels not in the correct label space result in **performance decreases of up to 16% absolute** in **direct models**

Seeing correct label space is important



Labels not in the correct label space result in **performance decreases of up to 2% absolute in channel models**

Format



Evaluate the importance of pairing an input sentence with a label

Changing the input-label format

*Demos
w/o labels*

(Format ✗ Input distribution ✓ Label space ✗ Input-label mapping ✗)
Circulation revenue has increased by 5% in Finland and 4% in Sweden in 2008.
Panostaja did not disclose the purchase price.

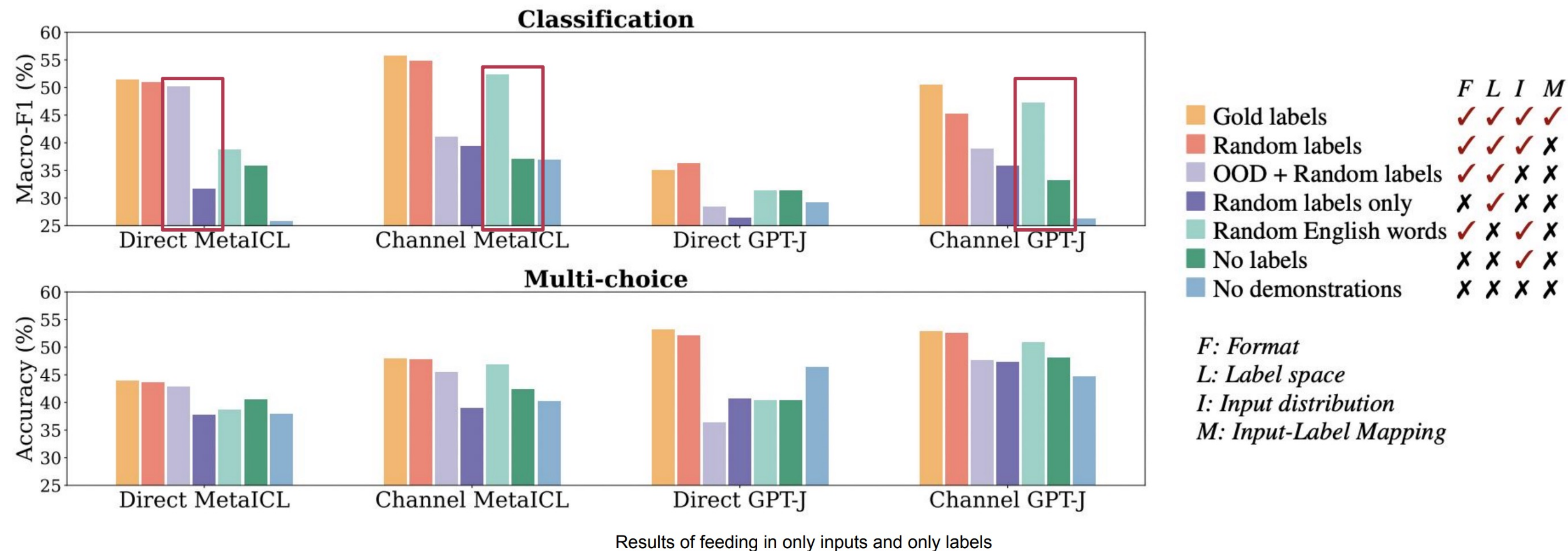
*Demos
labels only*

(Format ✗ Input distribution ✗ Label space ✓ Input-label mapping ✗)
positive
neutral

Examples with only inputs (top) and only labels (bottom)

Feed in examples with **no labels** and **with labels only**

Keeping the input-label format for demonstrations is vital for performance



Using **out-of-distribution inputs** and **random English words** as labels is better than only keeping **one part of the format** or having no demonstrations

Rethinking the Role of Demonstrations: Summary

انگ، Mapping، حسی نام نہی

- Having correct input-output pairs do not matter as much as long as we know the correct label space.
- Retaining the format (input-output pairs) whether by using (OOD + random labels) or (in-dist sentences + random English words) also decently improves performance.
- Connection to the Bayesian inference framework
 - all the components of the prompt are providing “evidence” to enable the model to better infer (locate) concepts that are learned during pretraining.

سم نو دیسی ترین ...

However...

Training examples (truncated)

```
beet: sport  
golf: animal  
horse: plant/vegetable  
corn: sport  
football: animal
```



Test input and predictions

```
monkey: plant/vegetable ✓  
panda: plant/vegetable ✓  
cucumber: sport ✓  
peas: sport ✓  
baseball: animal ✓  
tennis: animal ✓
```

An example synthetic task with unusual semantics that GPT-3 can successfully learn. A modified figure from Rong.

More directions to explore

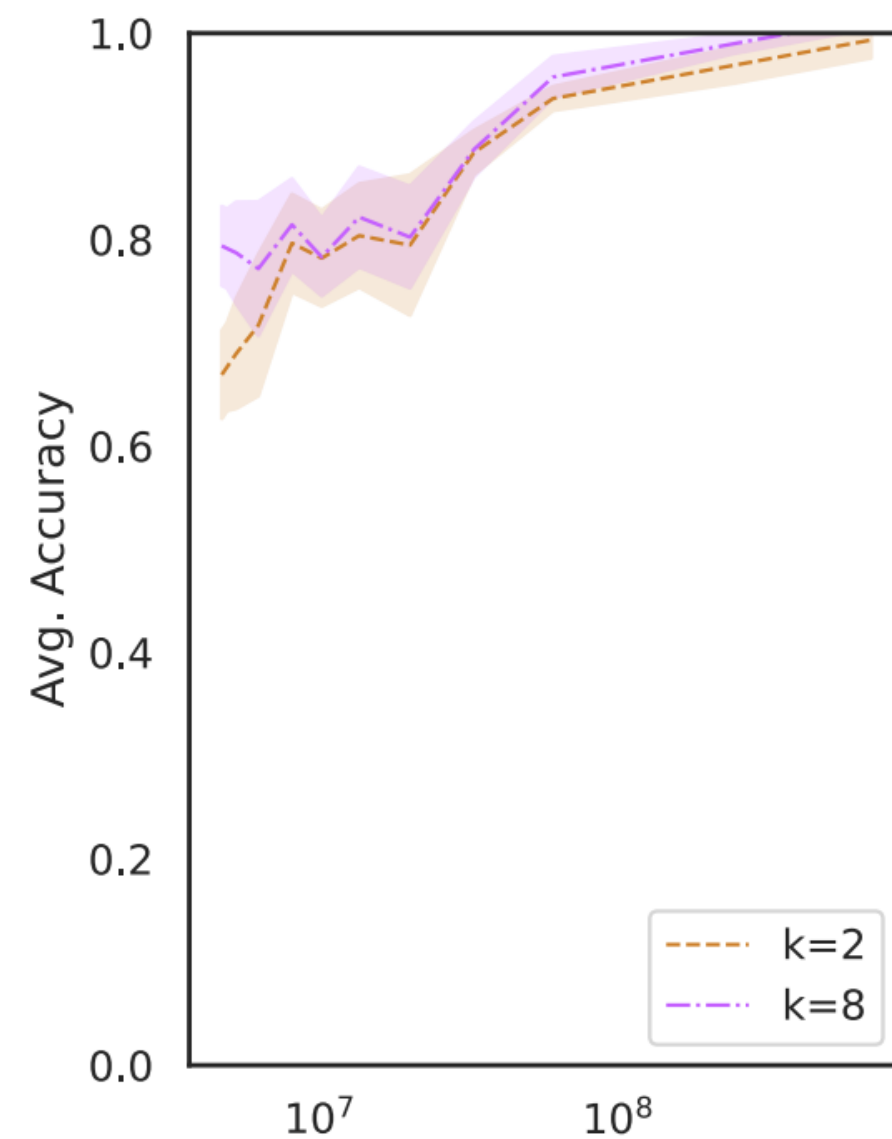
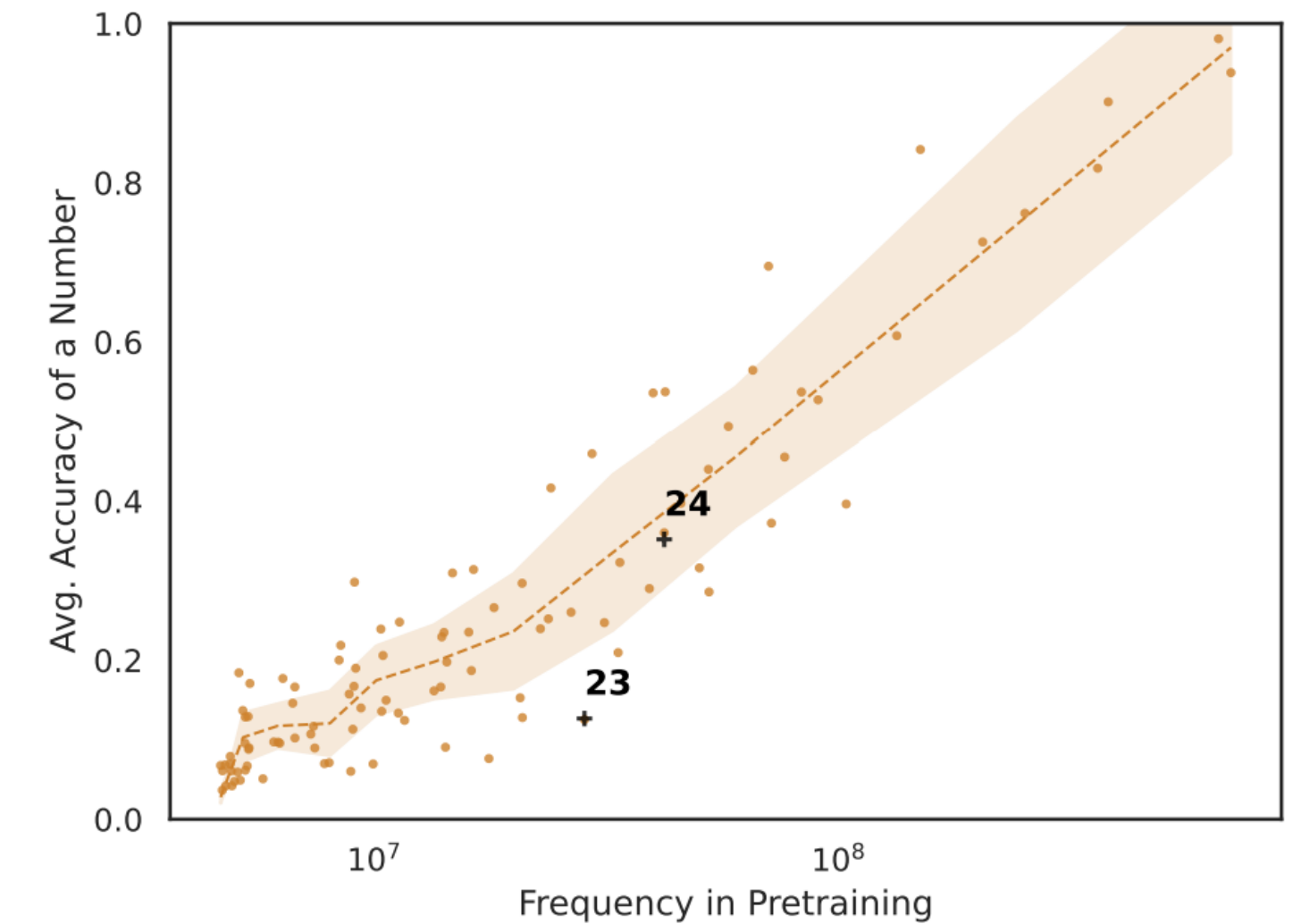
- **Understanding pre-training data** for in-context learning.
- Understanding model performance on “**unseen**” tasks
- Extending the framework to incorporate **task descriptions** as part of the prompts
- Capturing effects from **model architecture** and training
- **Variable length demonstrations**
 - i.e. k is different in each example

Pretraining data

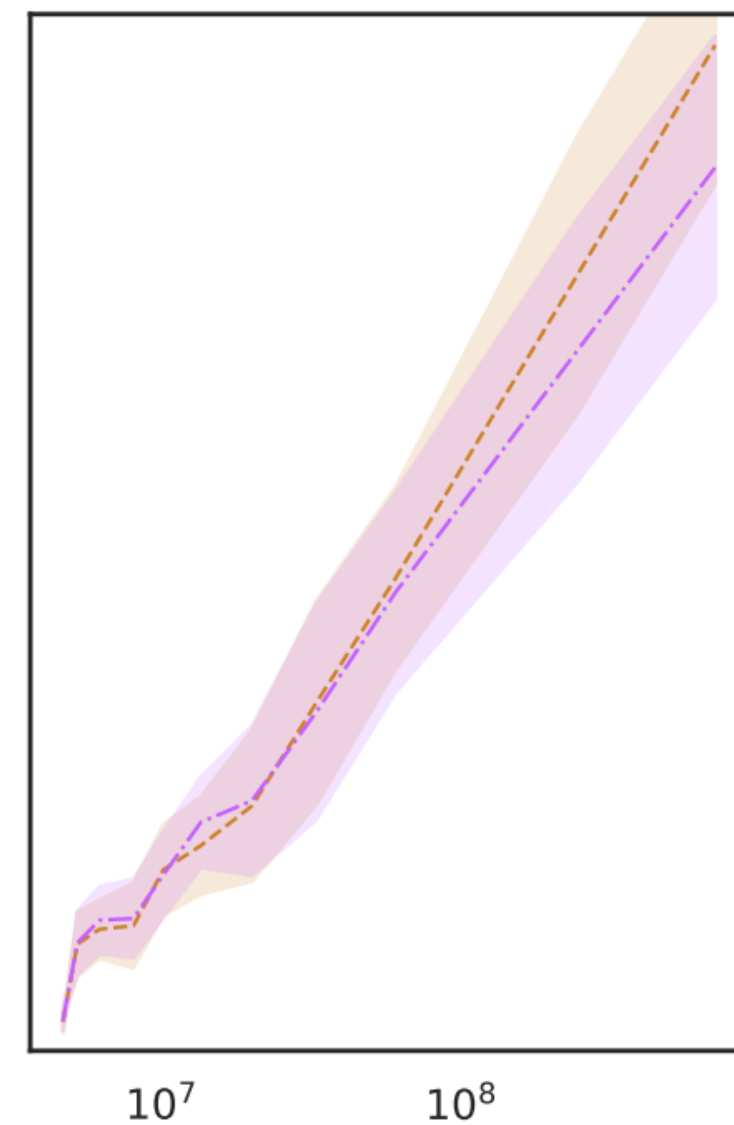
سوال 24 به از 23
جواب 24 است نه 23

Q: What is 24 times 18? Q: What is 23 times 18?
A: — Model: 432 ✓ A: — Model: 462 ✗

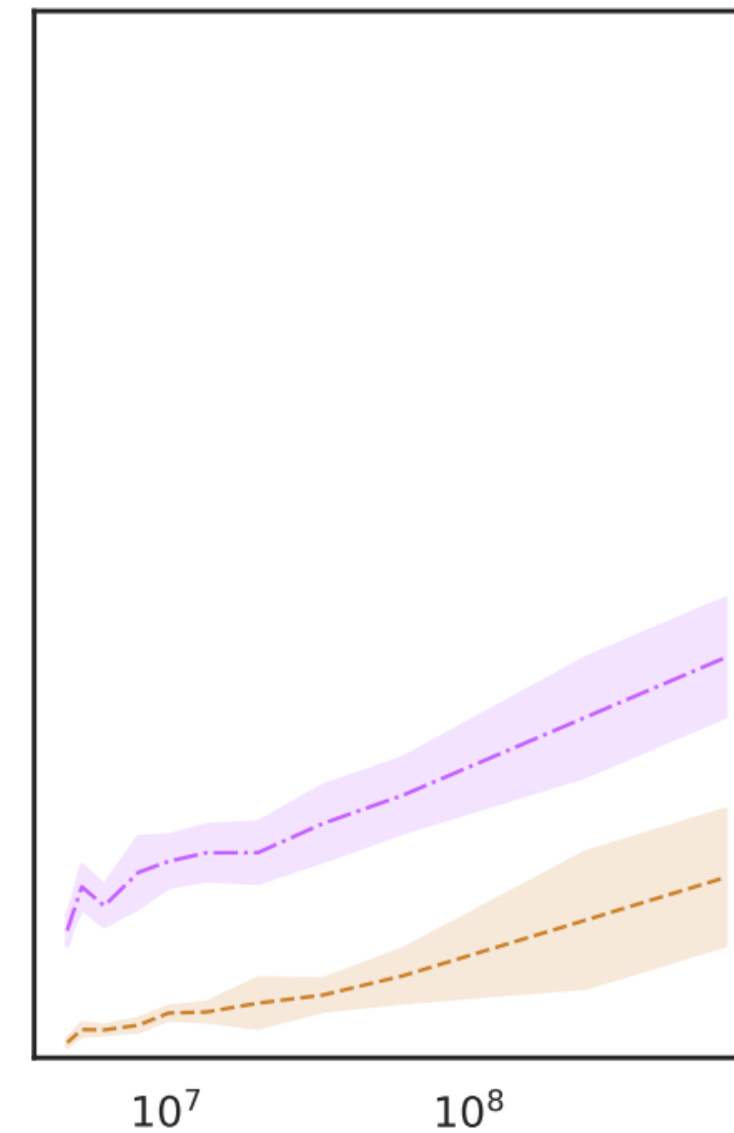
- In-context learning performance is highly correlated with term frequencies during pretraining



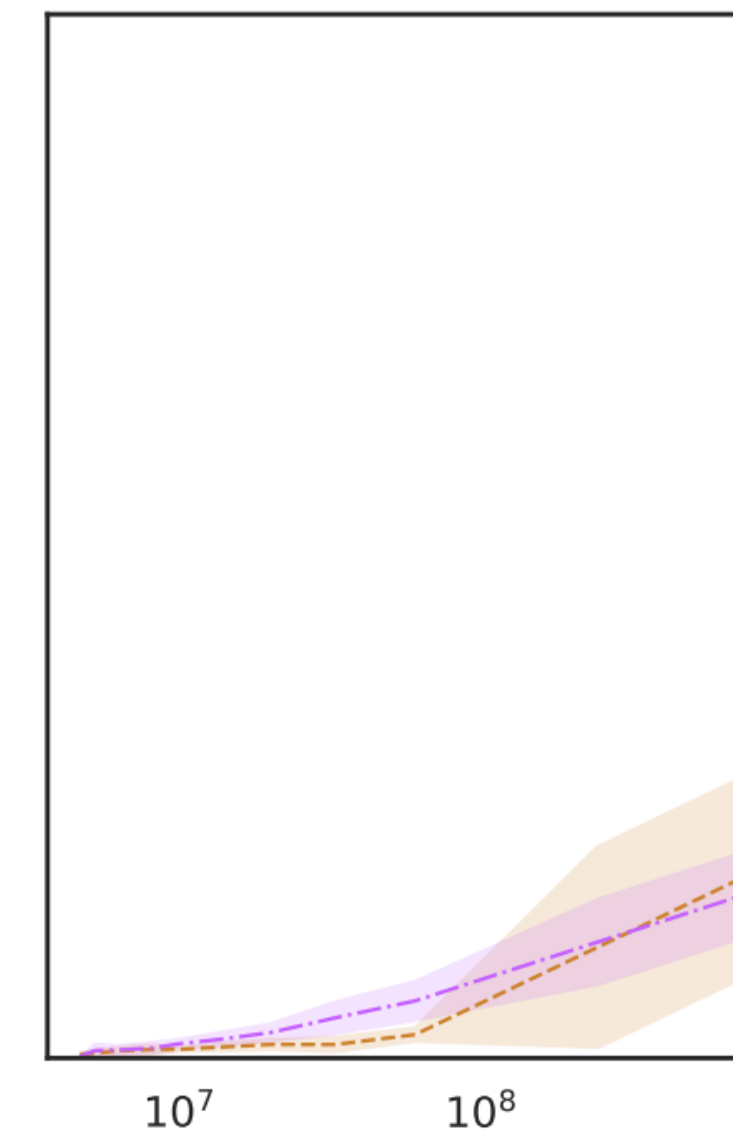
(a) Arithmetic-Addition



(b) Arithmetic-Multiplication



(c) Op.Inference-Addition



(d) Op. Inference-Mult.

Prompt design

- ICL is inherently unstable
- A surge of methods that search for robust and high-performing prompts:
 - Template search
 - all these methods require a high-quality validation set to do prompt selection or optimization
 - In-context example search

Choice of demonstrations (examples)

- ICL is powerful and versatile
- However, its performance depends heavily on the choice of examples.
- How to better select in-context examples?
 - retrieve examples that are semantically-similar to a test sample to formulate its corresponding prompt.

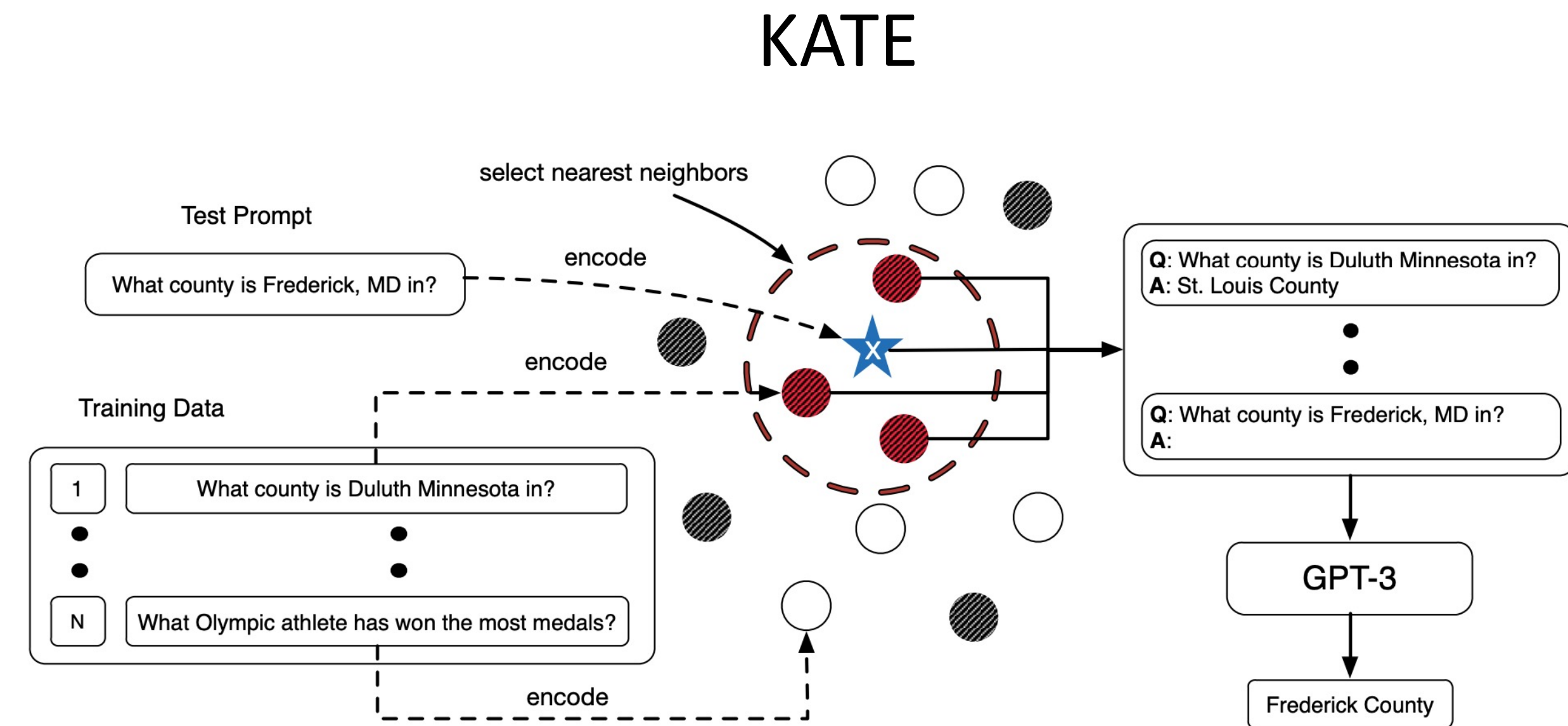
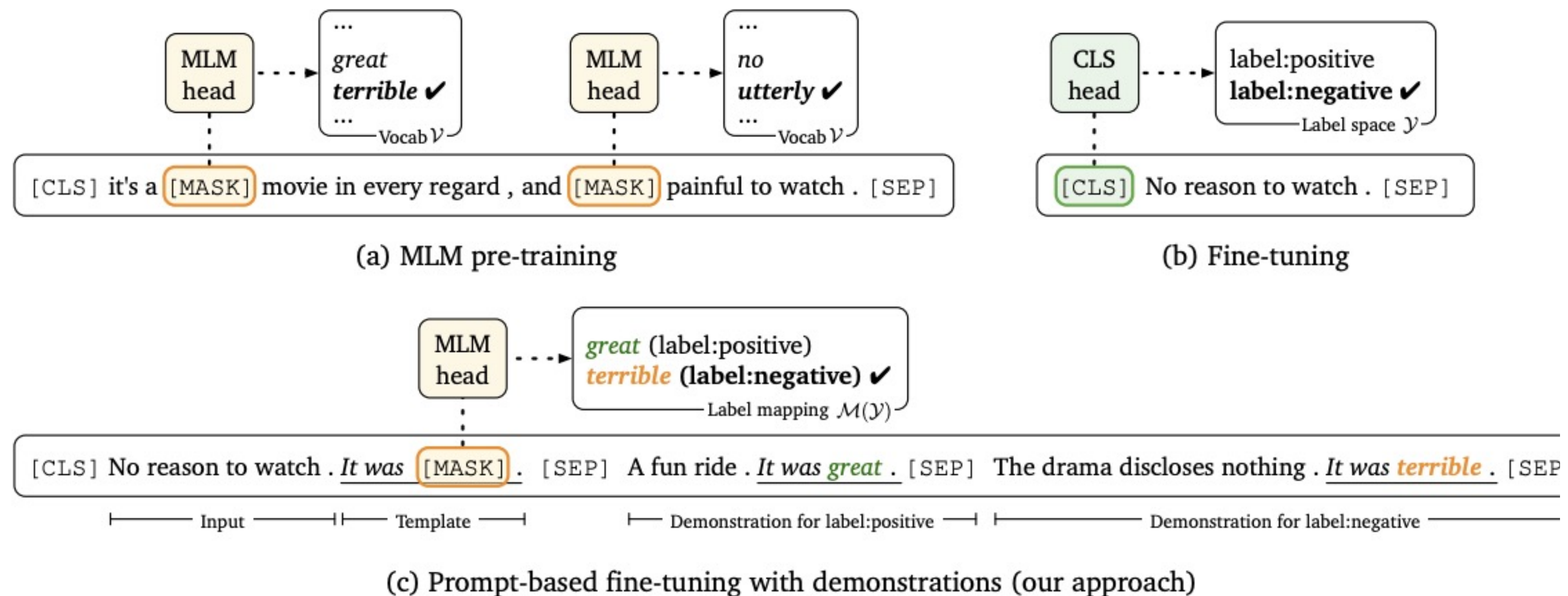


Figure 2: In-context example selection for GPT-3. White dots: unused training samples; grey dots: randomly sampled training samples; red dots: training samples selected by the k -nearest neighbors algorithm in the embedding space of a sentence encoder.

Recap: Other few-shot learning methods

- Prompt-based finetuning by verbalizers (e.g. LM-BFF)
 - LM-BFF re-uses the pre-trained weights and does not introduce any new parameters. It reduces the gap between pretraining and fine-tuning



Ways to adapt to new tasks

- Zero-shot learning
 - by task description through Prompt
 - T0: Multi-task training for zero-shot performance
- Few-shot learning
 - In-context learning
 - Verbalizer (i.e. a label word mapping)
- Lightweight Fine-tuning
 - Prompt tuning (Lester et al., 2021)
 - Prefix tuning (Li and Liang, 2021)
 - Adapter (Houlsby et al. 2019) and LoRA (Hu et al., 2021)
- Fine-tuning for human-aligned language models (later in the course)

Questions

